MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BUFFER, BIDIRECTIONAL, DIGITAL, KEY CODE HRL

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic bidirectional buffer module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

*2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.092 failures per million operating hours for class I or 0.253 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be HRL.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/287-1 for class I and M28787/287-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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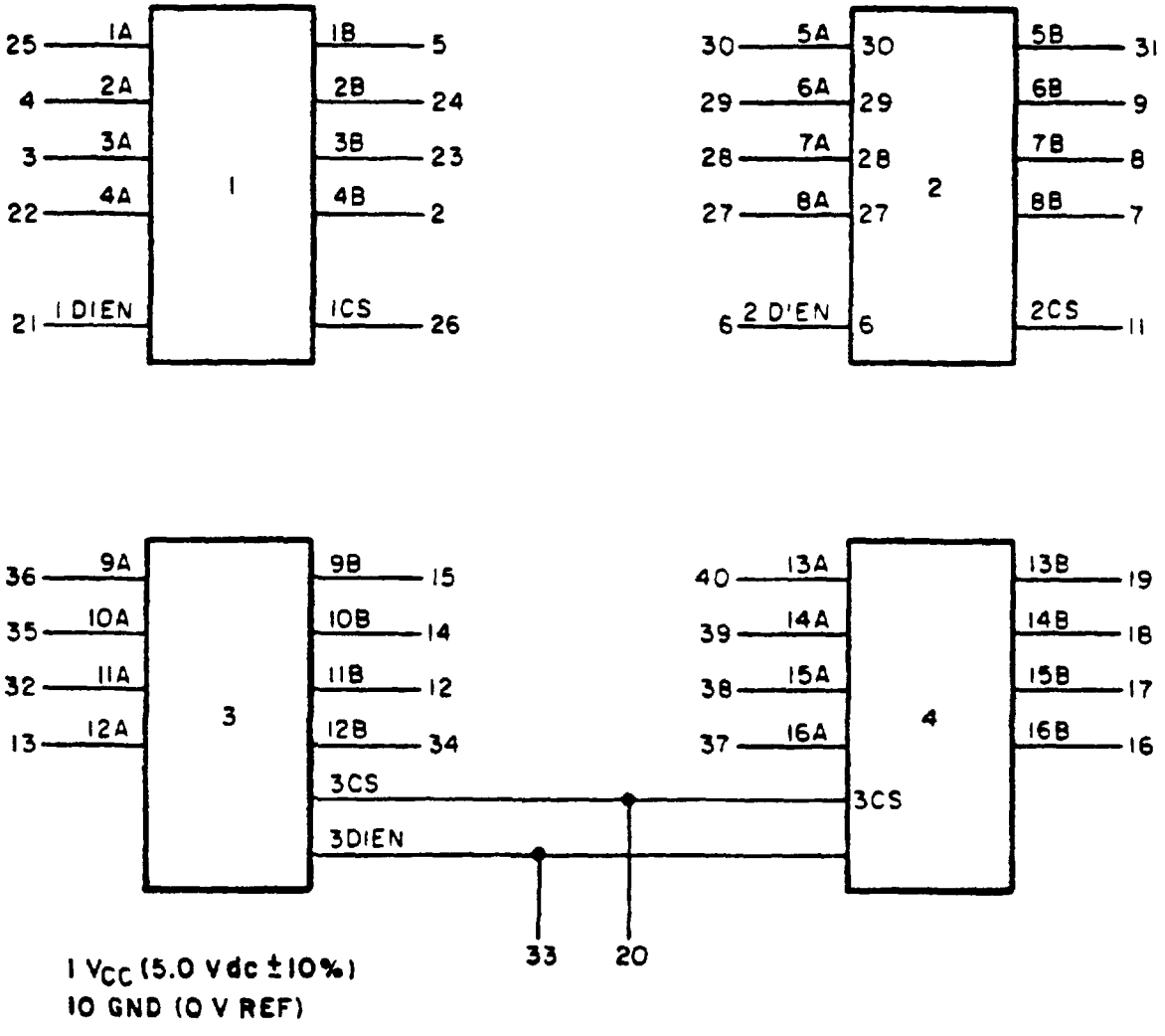


FIGURE 1. HRL module function diagram, bidirectional buffer.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} maximum stress	IV	Q	◄	----	Module must pass remaining tests							►	
C1-C16	Output short circuit current ("A" outputs)	V	X	-15	-65	-15	-65	-15	-65	-15	-65	-15	-65	mA
C17-C32	Output short circuit current ("B" outputs)	V	X	-30	-120	-30	-120	-30	-120	-30	-120	-30	-120	mA
D1-D38	Input clamp diode voltage	VI	X	0.0	-1.2	0.0	-1.2	0.0	-1.2	0.0	-1.2	0.0	-1.2	V
E1-E48	Output leakage current high level ("B" outputs)	VII	X	0	100	0	100	0	100	0	100	0	100	µA
E49-E96	Output leakage current high level ("A" outputs)	VII	X	0	60	0	60	0	60	0	60	0	60	µA
F1-F16	Output leakage current power off	VIII	X	0	60	0	60	0	60	0	60	0	60	µA
G1-G48	Output leakage current low level ("B" outputs)	IX	X	-250	100	-250	100	-250	100	-250	100	-250	100	µA
G49-G96	Output leakage current low level ("A" outputs)	IX	X	-250	20	-250	20	-250	20	-250	20	-250	20	µA
H1-H16	Output voltage, low level ("B" outputs), I_{OL} = 25 mA	X	X	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	V
H17-H32	Output voltage, low level ("A" outputs), I_{OL} = 15 mA	X	X	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	V
H33-H48	Output voltage, low level ("B" outputs), I_{OL} = 45 mA	X	X	0.00	0.60	0.00	0.60	0.00	0.60	0.00	0.60	0.00	0.60	V
J1-J16	Output voltage, high level ("A" outputs), I_{OH} = -0.5 mA	XI	X	3.4	4.5	3.4	4.5	3.4	4.5	3.4	4.5	3.4	4.5	V

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
J17-J32	Output voltage, high level ("B" outputs), I_{OH} = -5 mA	XI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K4	Input current, high level	XII	X	0	80	0	80	0	80	0	80	0	80	µA
K5,K6	Input current, high level	XII	X	0	160	0	160	0	160	0	160	0	160	µA
L1-L4	Input current, low level	XIII	X	-10	-550	-10	-550	-10	-550	-10	-550	-10	-550	µA
L5,L6	Input current, low level	XIII	X	-10	-1,100	-10	-1,100	-10	-1,100	-10	-1,100	-10	-1,100	µA
M	Power supply current	XIV	X	50	520	50	520	50	520	50	520	50	520	mA
N1-N16	Transition time (t_{TLH}), C_L = 50 pF ("B" outputs)	XV	S	2	25	2	25	2	27	2	25	2	30	ns
N17-N32	Transition time (t_{TLH}), C_L = 50 pF ("A" outputs)	XV	S	5	35	5	35	5	37	5	35	5	40	ns
P1-P16	Transition time (t_{THL}), C_L = 50 pF ("B" outputs)	XVI	S	2	10	2	10	2	11	2	10	2	13	ns
P17-P32	Transition time (t_{THL}), C_L = 50 pF ("A" outputs)	XVI	S	2	15	2	15	2	17	2	15	2	20	ns
Q1-Q16	Propagation delay time (t_{PLH}), C_L = 50 pF (A to B)	XVII	S	2	28	2	28	3	30	2	28	3	32	ns
Q17-Q32	Propagation delay time (t_{PLH}), C_L = 50 pF (B to A)	XVII	S	2	28	2	28	3	30	2	28	3	32	ns
Q33-Q48	Propagation delay time (t_{PLH}), C_L = 1000 pF (A to B)	XVII	Q	10	100	10	100	10	105	10	100	11	110	ns
R1-R16	Propagation delay time (t_{PHL}), C_L = 50 pF (A to B)	XVIII	S	2	28	2	28	3	30	2	28	3	32	ns
R17-R32	Propagation delay time (t_{PHL}), C_L = 50 pF (B to A)	XVIII	S	2	28	2	28	3	30	2	28	3	32	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R33-R48	Propagation delay time (t_{PHL}), C_L = 1000 pF (A to B)	XVIII	Q	10	100	10	100	10	105	10	100	11	110	ns
S1-S32	Output disable delay time (t_{OH}), C_L = 50 pF	XIX	S	4	45	4	45	5	50	4	45	5	55	ns
T1-T32	Output disable delay time (t_{HO}), C_L = 50 pF	XX	S	7	78	7	78	8	83	7	78	8	88	ns
U1-U16	Output disable delay time (t_{LH}), C_L = 50 pF ("B" outputs)	XXI	S	4	45	4	45	5	50	4	45	5	55	ns
U17-U32	Output disable delay time (t_{LH}), C_L = 50 pF ("A" outputs)	XXI	S	10	220	10	220	10	225	10	230	10	230	ns
V1-V32	Output disable delay time (t_{HL}), C_L = 50 pF	XXII	S	7	78	7	78	8	83	7	78	8	88	ns
W1-W32	Output crosstalk, low level	XXIII	Q	0	1,200	0	1,200	0	1,200	0	1,200	0	1,200	mV
X1-X32	Output crosstalk, high level	XXIV	Q	0	800	0	800	0	800	0	800	0	800	mV
Y	Power supply transient amplitude	XXV	Q	0	800	0	800	0	800	0	800	0	800	mV
Z	Power supply filter capacitance	XXVI	S	0.71	1.31	0.71	1.31	0.71	1.31	0.71	1.31	0.71	1.31	µF
AA1-AA32	Input capacitance	XXVII	Q	2	25	2	25	2	25	2	25	2	25	pF
AA33-AA36	Input capacitance	XXVII	Q	2	15	2	15	2	15	2	15	2	15	pF
AA37,AA38	Input capacitance	XXVII	Q	4	30	4	30	4	30	4	30	4	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through V and Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Z, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through V and Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through AA shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (V_{CC} = 5 V, outputs not loaded)	2.86	2.86	W
Input voltage			
Low level	0.95 max	0.95 max	V
High level	2.0 min	2.0 min	V

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Input current (1 and 2DIEN and 1 and 2CS inputs)			
Low level (V_{IL} = 0 V)	-550 max	-550 max	µA
Low level (V_{IL} = 0.45 V)	-500 max	-500 max	µA
High level (V_{IH} = 3.4 V)	80 max	80 max	µA
Input current (3DIEN and 3CS inputs)			
Low level (V_{IL} = 0 V)	-1,100 max	-1,100 max	µA
Low level (V_{IL} = 0.45 V)	-1,000 max	-1,000 max	µA
High level (V_{IH} = 2.4 V)	160 max	160 max	µA
Output voltage			
Low level (A outputs, I_{OL} = 15 mA)	0.45 max	0.45 max	V
(B outputs, I_{OL} = 25 mA)	0.45 max	0.45 max	V
(B outputs, I_{OL} = 45 mA)	0.60 max	0.60 max	V
High level (A outputs, I_{OH} = -0.5 mA)	3.4 min	3.4 min	V
(B outputs, I_{OH} = -5.0 mA)	2.4 min	2.4 min	V
Output leakage current			
High level (A outputs, V_{OH} = 3.4 V)	60 max	60 max	µA
(B outputs, V_{OH} = 2.4 V)	100 max	100 max	µA
Low level (A outputs, V_{OL} = 0.45 V)	-250 min	-250 min	µA
(A outputs, V_{OL} = 0.45 V)	20 max	20 max	µA
(B outputs, V_{OL} = 0.45 V)	-250 min	-250 min	µA
(B outputs, V_{OL} = 0.45 V)	100 max	100 max	µA
Propagation delay time (C_L = 50 pF)			
t_{PLH} (A to B and B to A)	3 min, 30 max	3 min, 32 max	ns
t_{PHL} (A to B and B to A)	3 min, 30 max	3 min, 32 max	ns
Propagation delay time (C_L = 1,000 pF)			
t_{PLH} (A to B)	10 min, 105 max	11 min, 110 max	ns
t_{PHL} (A to B)	10 min, 105 max	11 min, 110 max	ns
Propagation delay time (C_L = 50 pF			
Disable to output			
t_{IH} (B outputs)	4 min, 50 max	4 min, 55 max	ns
t_{IH} (A outputs)	10 min, 175 max	10 min, 230 max	ns
Disable to A and B outputs			
t_{HI}	7 min, 83 max	7 min, 88 max	ns
t_{OH}	4 min, 50 max	4 min, 55 max	ns
t_{HO}	7 min, 83 max	7 min, 88 max	ns

6.5 Module function. This module contains four 4 input three-state bidirectional bus drivers.

6.5.1 Input to output relationship. Each buffer has the input to output relationship of table II for positive logic.

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TABLE II. Input to output relationship. 1/ - 3/

Inputs				Outputs	
DIEN	CS	A	B	A	B
L	L	L	X	High impedance	L
L	L	H	X	High impedance	H
L	H	X	X	High impedance	High impedance
H	L	X	L	L	High impedance
H	L	X	H	H	High impedance
H	H	X	X	High impedance	High impedance

1/ L = Low level.
2/ H = High level.
3/ X = Don't care.

6.5.2 Parallel outputs Paralleling of outputs is not permissible.

6.6 User information. All "B" outputs of the HRL module have a diode from the output to V_{CC}. The bus that these outputs are connected to should be contained within one cabinet. If the bus is run between two cabinets and power is lost in one of the cabinets, the diodes in all the HRL modules will forward bias and pull the bus down. The "A" outputs do not have this diode.

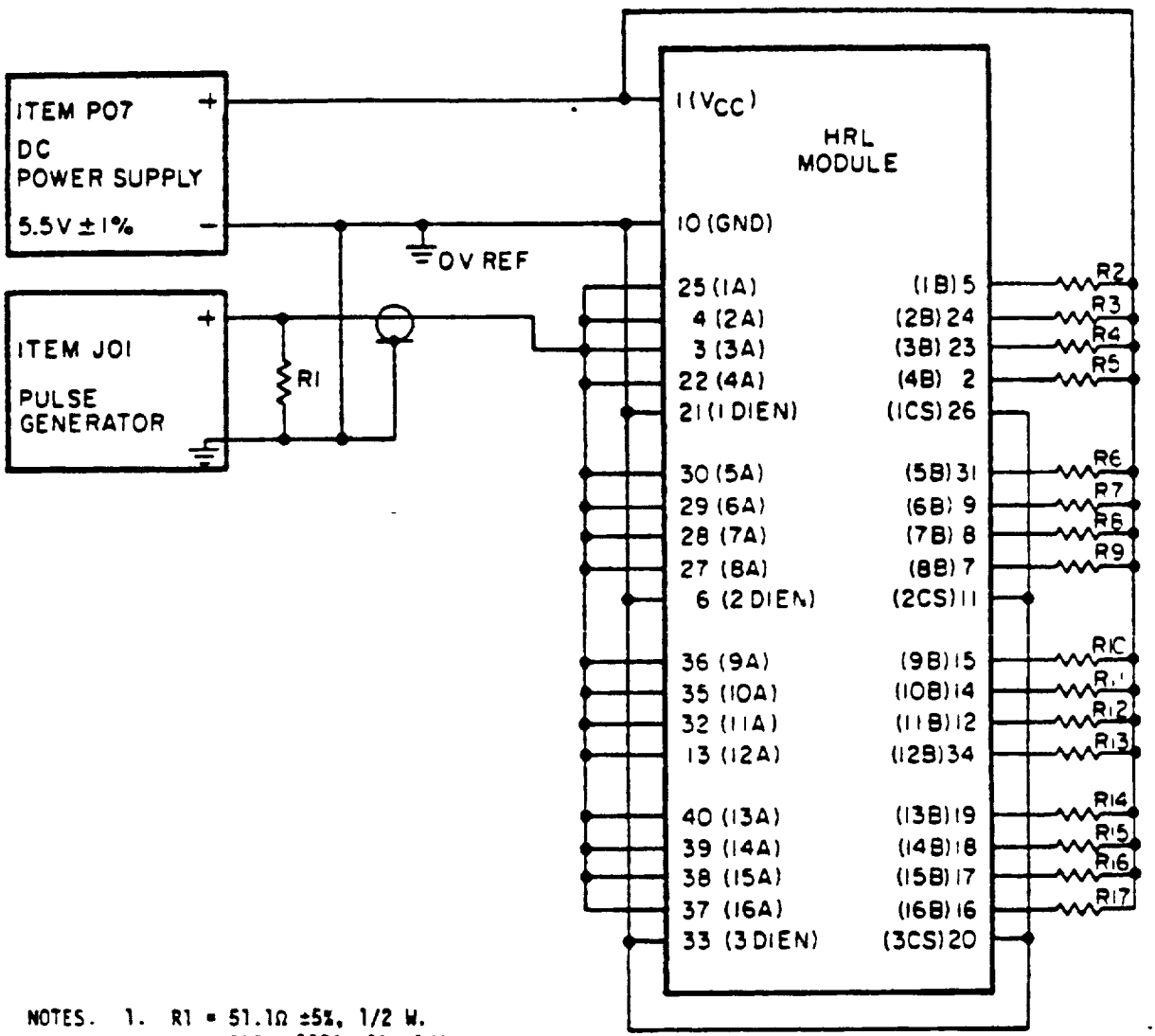
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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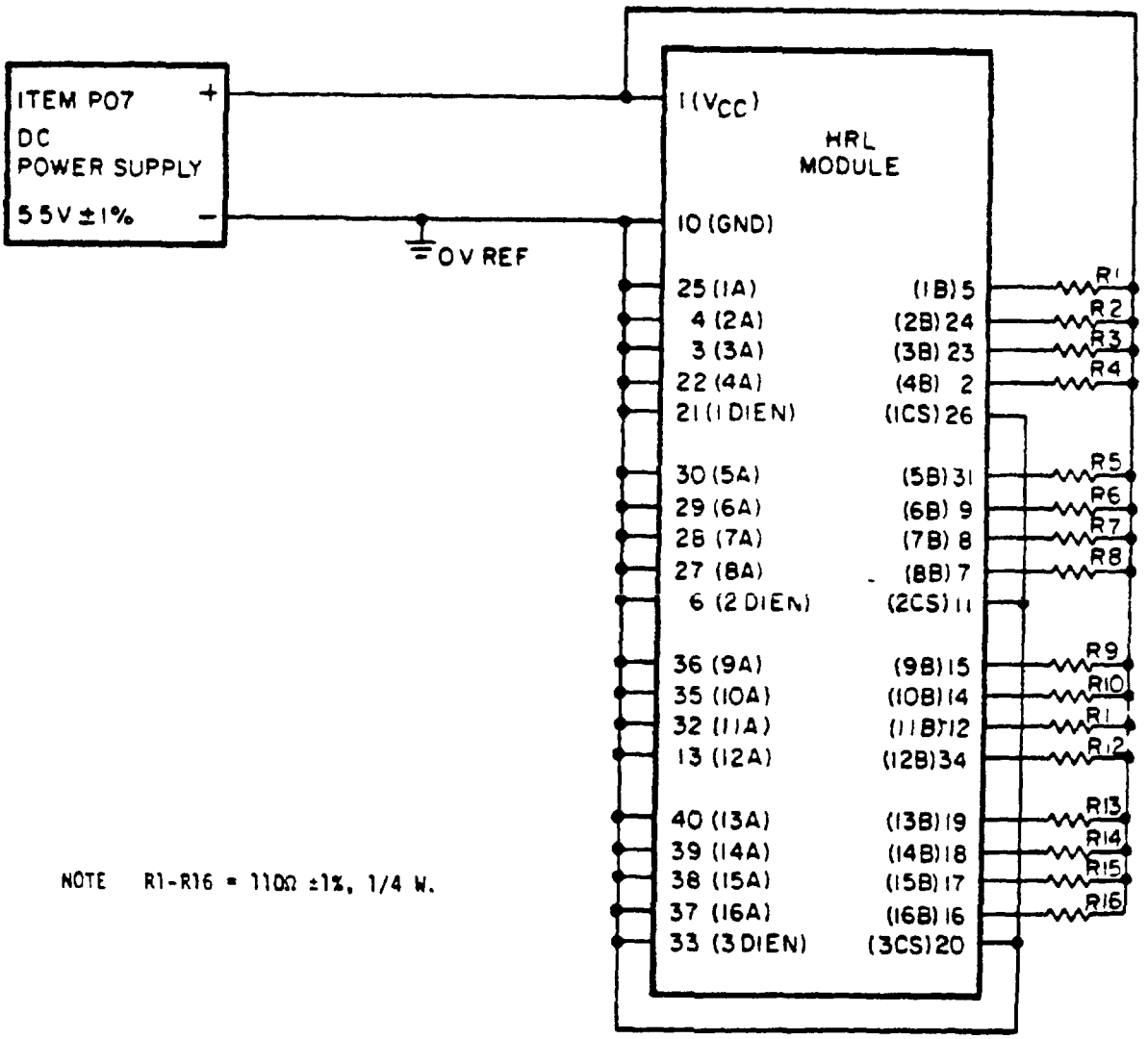
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NOTES. 1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 - R17 = 110Ω ±1%, 1/4 W.
3. Item J01 setup:
PW = 500 ±50 ns.
PRR = 1.0 MHz ±10%.
PA = 4.0 ±0.5 V.
$t_{TLH} = t_{THL}$ = 50 ±10 ns.

FIGURE 2 Operating temperature and life test circuit

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NOTE R1-R16 = 110Ω ±1%, 1/4 W.

FIGURE 3 Component temperature test circuit

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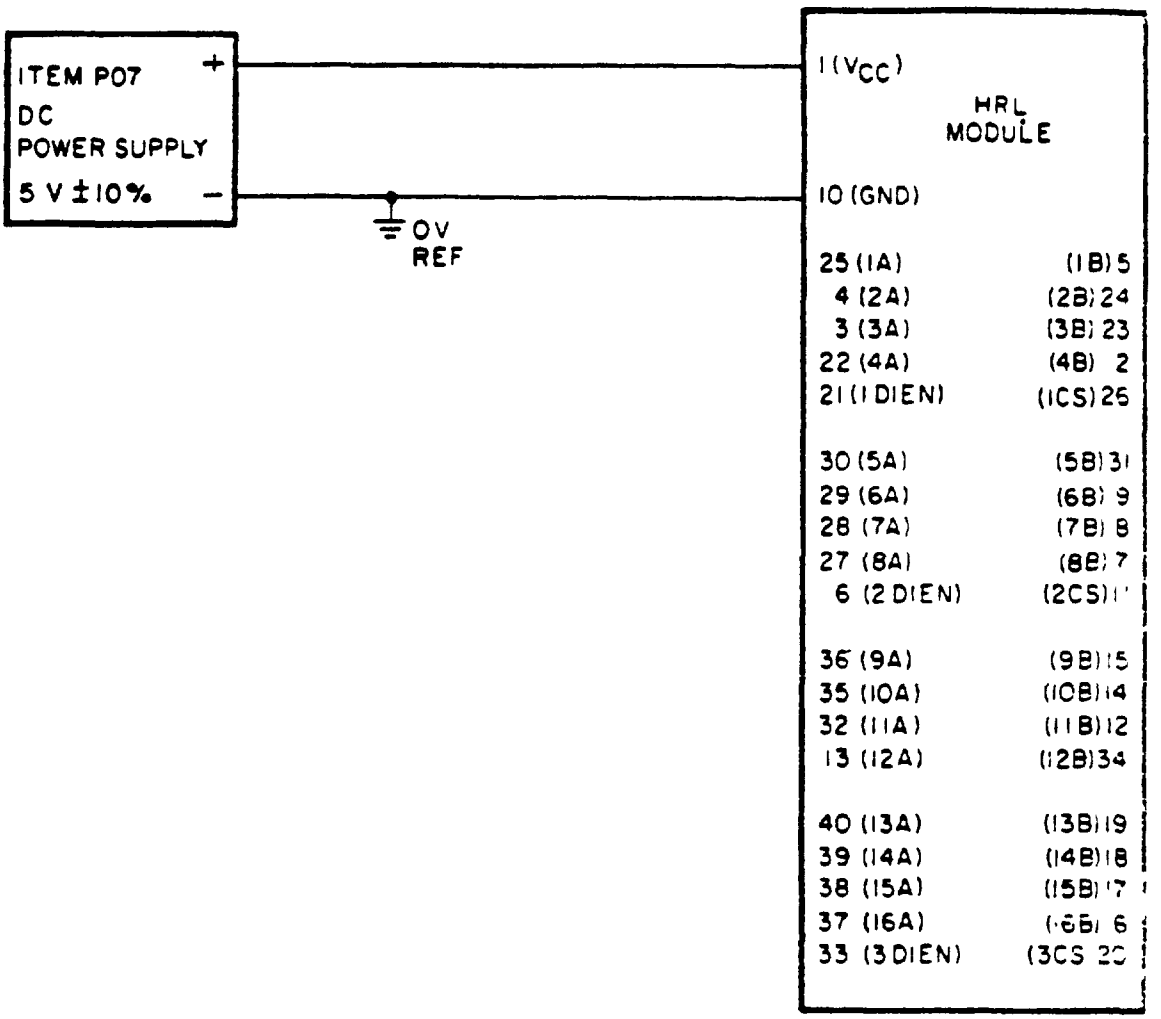


FIGURE 4. Vibration test circuit

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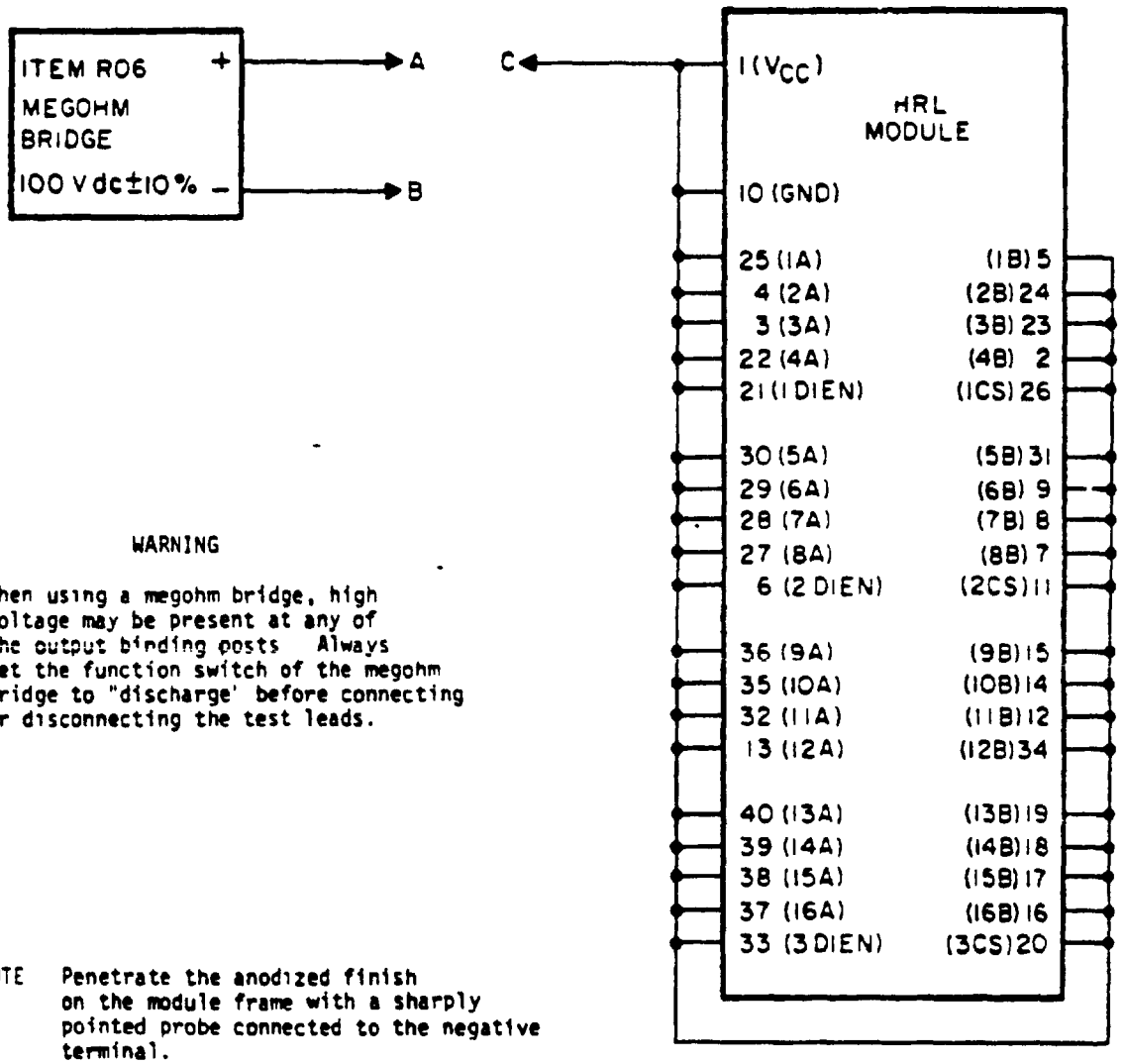
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	-	MΩ	5

TABLE IV. V_{CC} maximum stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
B	V_{CC} maximum stress	Module must pass remaining tests			6

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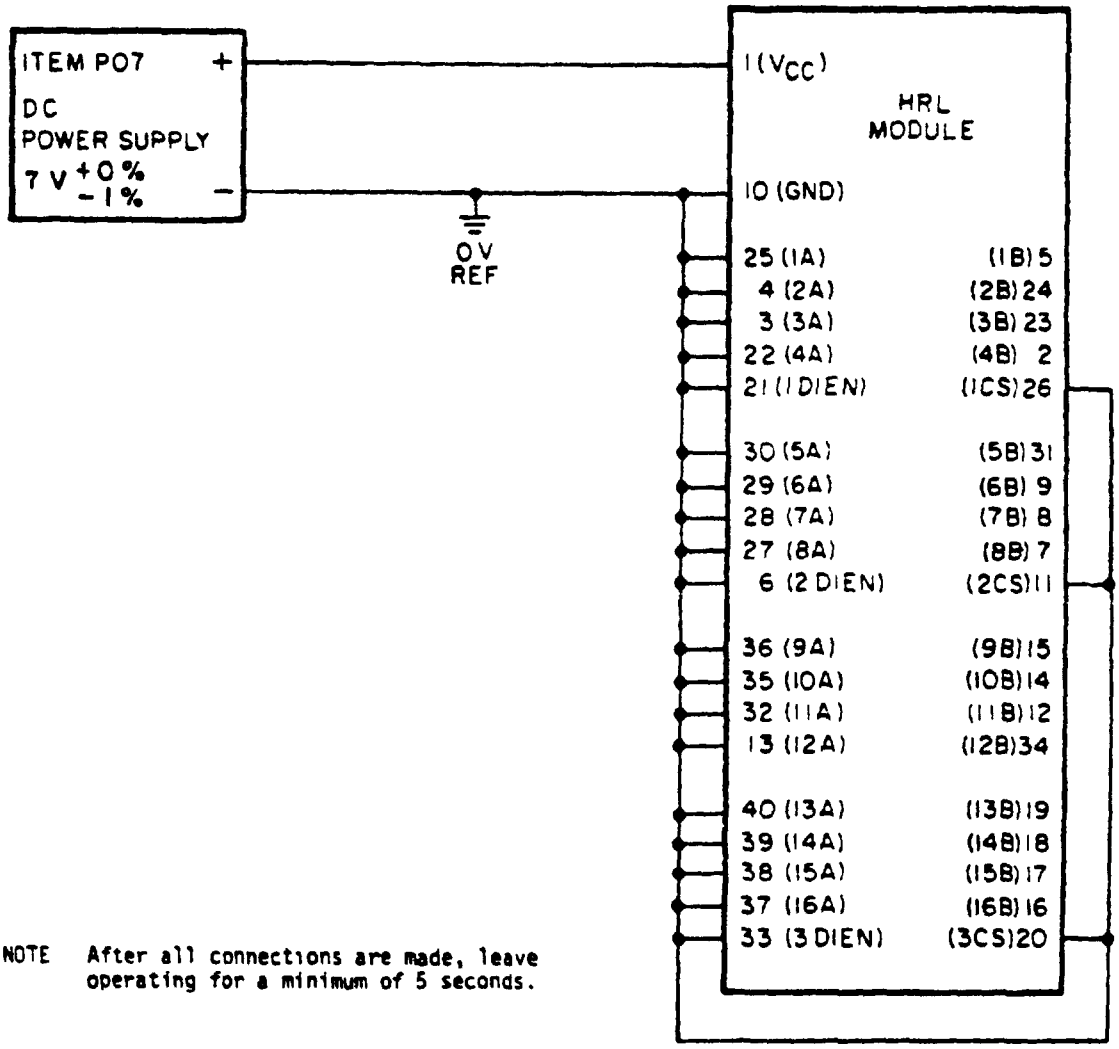
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the function switch of the megohm bridge to "discharge' before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5 Isolation test circuit for table III

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NOTE After all connections are made, leave operating for a minimum of 5 seconds.

FIGURE 6 V_{CC} maximum stress test circuit for table IV

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TABLE V. Output short circuit current test.

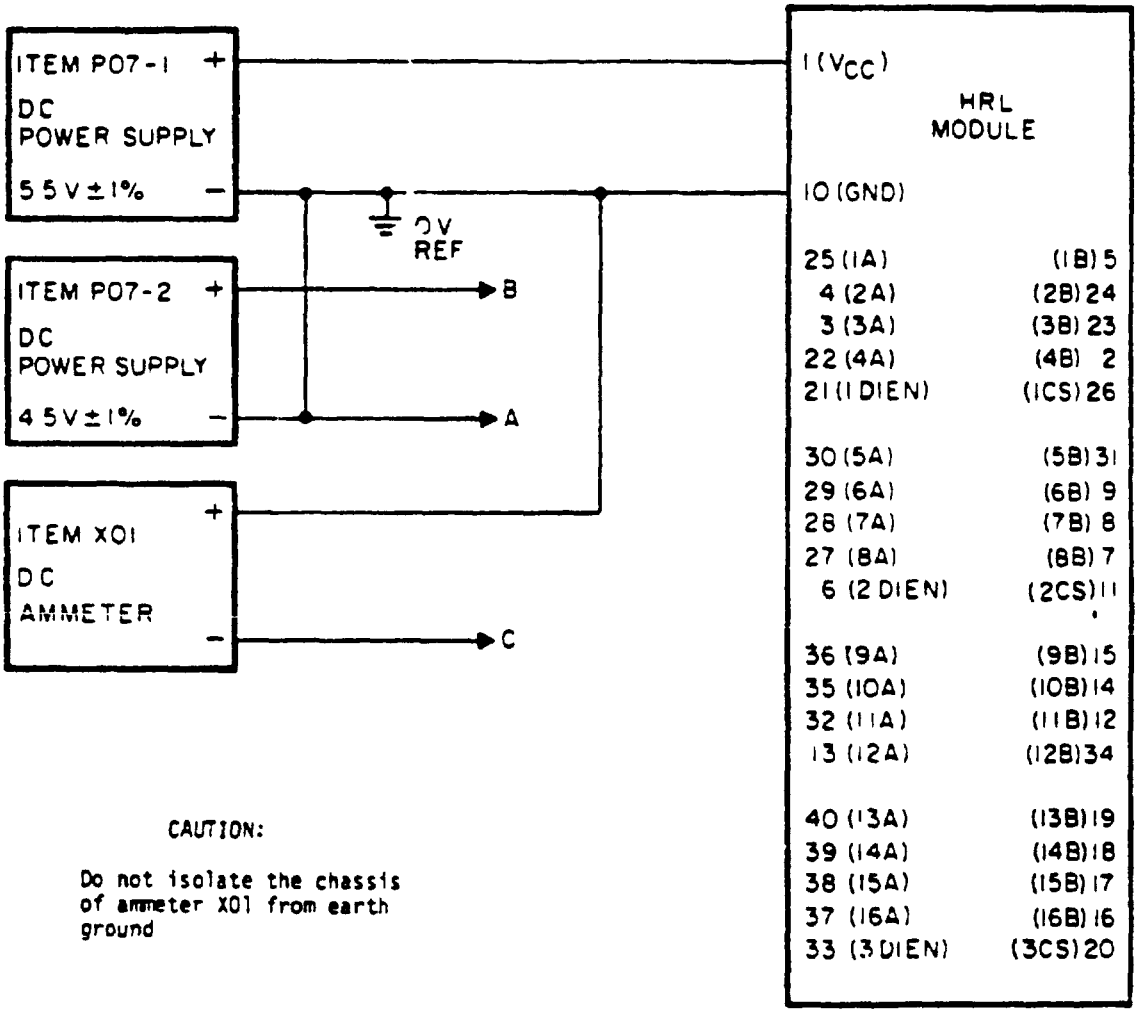
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	1A Output short circuit current	26,11,20	5,24,23,2,21,31,9,8,7,6,15,14, 12,34,19,18,17,16,33	25	-15	-65	mA	7
C2	2A "	"	"	4	"	"	"	"
C3	3A "	"	"	3	"	"	"	"
C4	4A "	"	"	22	"	"	"	"
C5	5A "	"	"	30	"	"	"	"
C6	6A "	"	"	29	"	"	"	"
C7	7A "	"	"	28	"	"	"	"
C8	8A "	"	"	27	"	"	"	"
C9	9A "	"	"	36	"	"	"	"
C10	10A "	"	"	35	"	"	"	"
C11	11A "	"	"	32	"	"	"	"
C12	12A "	"	"	13	"	"	"	"
C13	13A "	"	"	40	"	"	"	"
C14	14A "	"	"	39	"	"	"	"
C15	15A "	"	"	38	"	"	"	"
C16	16A "	"	"	37	"	"	"	"
C17	1B "	21,26,6, 11,33,20	25,4,3,22,30,29,28,27,36,35, 12,13,40,39,38,37	5	-30	-120	"	"
C18	2B "	"	"	24	"	"	"	"
C19	3B "	"	"	23	"	"	"	"
C20	4B "	"	"	2	"	"	"	"
C21	5B "	"	"	31	"	"	"	"

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TABLE V. Output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C22	6B Output short circuit current	21,26,6 11,33,20	25,4,3,22,30,29,28,27,36,35, 32,13,40,19,38,37,	9	-30	-120	mA	7
C23	7B "	"	"	8	"	"	"	"
C24	8B "	"	"	7	"	"	"	"
C25	9B "	"	"	15	"	"	"	"
C26	10B "	"	"	14	"	"	"	"
C27	11B "	"	"	12	"	"	"	"
C28	12B "	"	"	34	"	"	"	"
C29	13B "	"	"	19	"	"	"	"
C30	14B "	"	"	18	"	"	"	"
C31	15B "	"	"	17	"	"	"	"
C32	16B "	"	"	16	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground

NOTE Test duration shall not exceed 5 seconds.

FIGURE 7 Output short circuit current test circuit for table V

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TABLE VI. Input clamp diode voltage test.

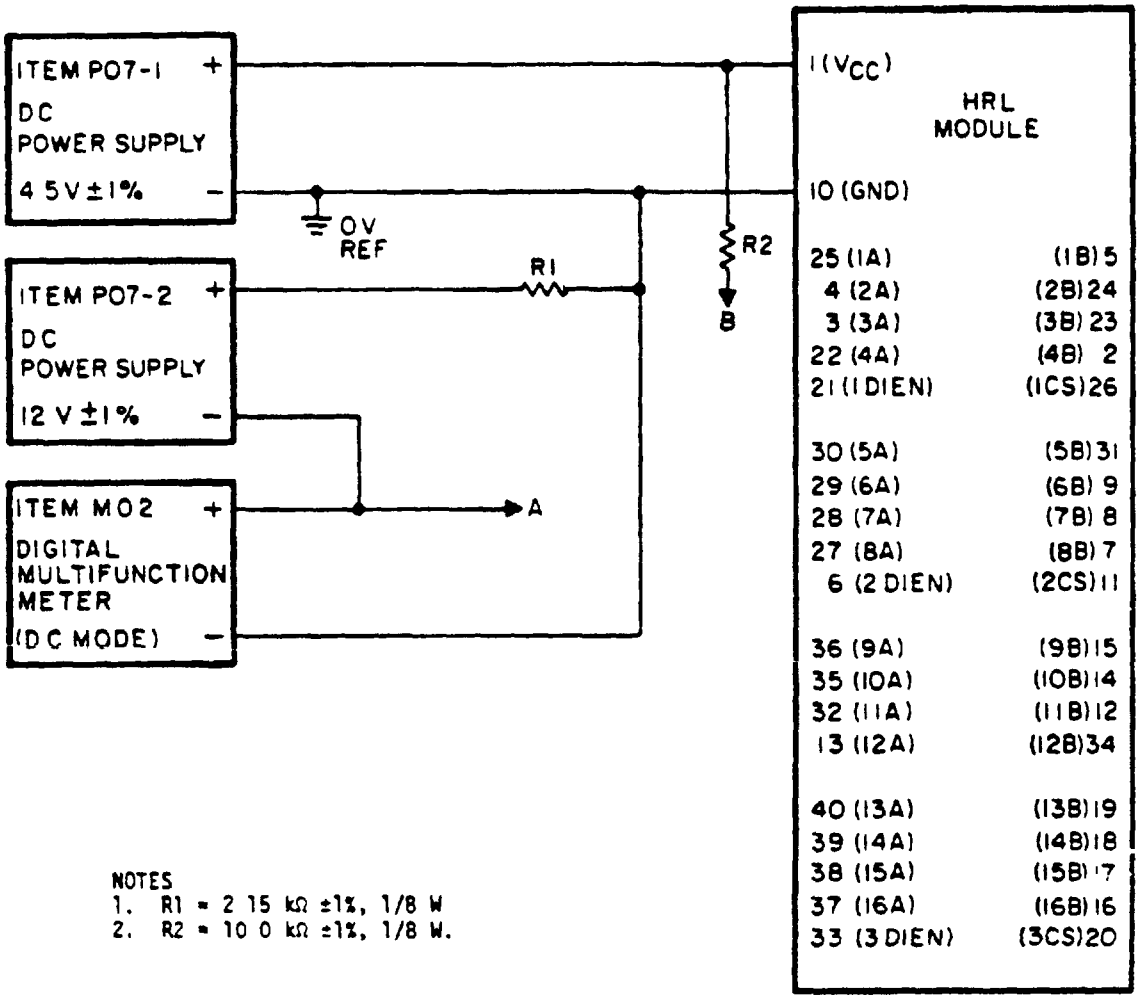
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1DIEN Input clamp diode voltage	21	Open	0.0	-1.2	V	8
D2	1CS "	26	"	"	"	"	"
D3	2DIEN "	6	"	"	"	"	"
D4	2CS "	11	"	"	"	"	"
D5	3DIEN "	33	"	"	"	"	"
D6	3CS "	20	"	"	"	"	"
D7	1A "	25	21,26,6,11, 33,20	"	"	"	"
D8	2A "	4	"	"	"	"	"
D9	3A "	3	"	"	"	"	"
D10	4A "	22	"	"	"	"	"
D11	5A "	30	"	"	"	"	"
D12	6A "	29	"	"	"	"	"
D13	7A "	28	"	"	"	"	"
D14	8A "	27	"	"	"	"	"
D15	9A "	36	"	"	"	"	"
D16	10A "	35	"	"	"	"	"
D17	11A "	32	"	"	"	"	"
D18	12A "	13	"	"	"	"	"
D19	13A "	40	"	"	"	"	"
D20	14A "	39	"	"	"	"	"
D21	15A "	38	"	"	"	"	"

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TABLE VI. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D22	16A Input clamp diode voltage	37	21,26,6,11, 33,20	0.0	-1.2	V	8
D23	1B "	5	"	"	"	"	"
D24	2B "	24	"	"	"	"	"
D25	3B "	23	"	"	"	"	"
D26	4B "	2	"	"	"	"	"
D27	5B "	31	"	"	"	"	"
D28	6B "	9	"	"	"	"	"
D29	7B "	8	"	"	"	"	"
D30	8B "	7	"	"	"	"	"
D31	9B "	15	"	"	"	"	"
D32	10B "	14	"	"	"	"	"
D33	11B "	17	"	"	"	"	"
D34	12B "	34	"	"	"	"	"
D35	13B "	19	"	"	"	"	"
D36	14B "	18	"	"	"	"	"
D37	15B "	17	"	"	"	"	"
D38	16B "	16	"	"	"	"	"

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NOTES
1. R1 = 2 15 kΩ ±1%, 1/8 W
2. R2 = 10 0 kΩ ±1%, 1/8 W.

* FIGURE 8. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Output leakage current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	1B Output leakage current, high level	21,26	25	5	Open	0	100	µA	9
E2	2B "	"	4	24	"	"	"	"	"
E3	3B "	"	3	23	"	"	"	"	"
E4	4B "	"	22	2	"	"	"	"	"
E5	5B "	6,11	30	31	"	"	"	"	"
E6	6B "	"	29	9	"	"	"	"	"
E7	7B "	"	28	8	"	"	"	"	"
E8	8B "	"	27	7	"	"	"	"	"
E9	9B "	33,20	36	15	"	"	"	"	"
E10	10B "	"	35	14	"	"	"	"	"
E11	11B "	"	32	12	"	"	"	"	"
E12	12B "	"	13	34	"	"	"	"	"
E13	13B "	"	40	19	"	"	"	"	"
E14	14B "	"	39	18	"	"	"	"	"
E15	15B "	"	38	17	"	"	"	"	"
E16	16B "	"	37	16	"	"	"	"	"
E17	1B "	21	Open	5	26	"	"	"	"
E18	2B "	"	"	24	"	"	"	"	"
E19	3B "	"	"	23	"	"	"	"	"
E20	4B "	"	"	2	"	"	"	"	"

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TABLE VII. Output leakage current (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E21	5B Output leakage current, high level	6	Open	31	11	0	100	µA	9
E22	6B "	"	"	9	"	"	"	"	"
E23	7B "	"	"	8	"	"	"	"	"
E24	8B "	"	"	7	"	"	"	"	"
E25	9B "	33	"	15	20	"	"	"	"
E26	10B "	"	"	14	"	"	"	"	"
E27	11B "	"	"	12	"	"	"	"	"
E28	12B "	"	"	34	"	"	"	"	"
E29	13B "	"	"	19	"	"	"	"	"
E30	14B "	"	"	18	"	"	"	"	"
E31	15B "	"	"	17	"	"	"	"	"
E32	16B "	"	"	16	"	"	"	"	"
E33	1B "	26	25	5	21	"	"	"	"
E34	2B "	"	4	24	"	"	"	"	"
E35	3B "	"	3	23	"	"	"	"	"
E36	4B "	"	22	2	"	"	"	"	"
E37	5B "	11	30	31	6	"	"	"	"
E38	6B "	"	29	9	"	"	"	"	"
E39	7B "	"	28	8	"	"	"	"	"
E40	8B "	"	27	7	"	"	"	"	"

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TABLE VII. Output leakage current (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E41	9B Output leakage current, high level	20	36	15	33	0	100	µA	9
E42	10B "	"	35	14	"	"	"	"	"
E43	11B "	"	32	12	"	"	"	"	"
E44	12B "	"	13	34	"	"	"	"	"
E45	13B "	"	40	19	"	"	"	"	"
E46	14B "	"	39	18	"	"	"	"	"
E47	15B "	"	38	17	"	"	"	"	"
E48	16B "	"	37	16	"	"	"	"	"
E49	1A "	Open	Open	25	21,26	"	60	"	"
E50	2A "	"	"	4	"	"	"	"	"
E51	3A "	"	"	3	"	"	"	"	"
E52	4A "	"	"	22	"	"	"	"	"
E53	5A "	"	"	30	6,11	"	"	"	"
E54	6A "	"	"	29	"	"	"	"	"
E55	7A "	"	"	28	"	"	"	"	"
E56	8A "	"	"	27	"	"	"	"	"
E57	9A "	"	"	36	33,20	"	"	"	"
E58	10A "	"	"	35	"	"	"	"	"
E59	11A "	"	"	32	"	"	"	"	"
E60	12A "	"	"	13	"	"	"	"	"
E61	13A "	"	"	40	"	"	"	"	"

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TABLE VII. Output leakage current (high-level) test - Continued

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E62	14A Output leakage current, high level	Open	Open	39	33,20	0	60	uA	9
E63	15A "	"	"	38	"	"	"	"	"
E64	16A "	"	"	37	"	"	"	"	"
E65	1A "	26	5	25	21	"	"	"	"
E66	2A "	"	24	4	"	"	"	"	"
E67	3A "	"	23	3	"	"	"	"	"
E68	4A "	"	2	22	"	"	"	"	"
E69	5A "	11	31	30	6	"	"	"	"
E70	6A "	"	9	29	"	"	"	"	"
E71	7A "	"	8	28	"	"	"	"	"
E72	8A "	"	7	27	"	"	"	"	"
E73	9A "	20	15	36	33	"	"	"	"
E74	10A "	"	14	35	"	"	"	"	"
E75	11A "	"	12	32	"	"	"	"	"
E76	12A "	"	34	13	"	"	"	"	"
E77	13A "	"	19	40	"	"	"	"	"
E78	14A "	"	18	39	"	"	"	"	"
E79	15A "	"	17	38	"	"	"	"	"
E80	16A "	"	16	37	"	"	"	"	"
E81	1A "	21,26	5	25	Open	"	"	"	"
E82	2A "	21,26	24	4	Open	"	"	"	"

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TABLE VII. Output leakage current (high-level) test - Continued.

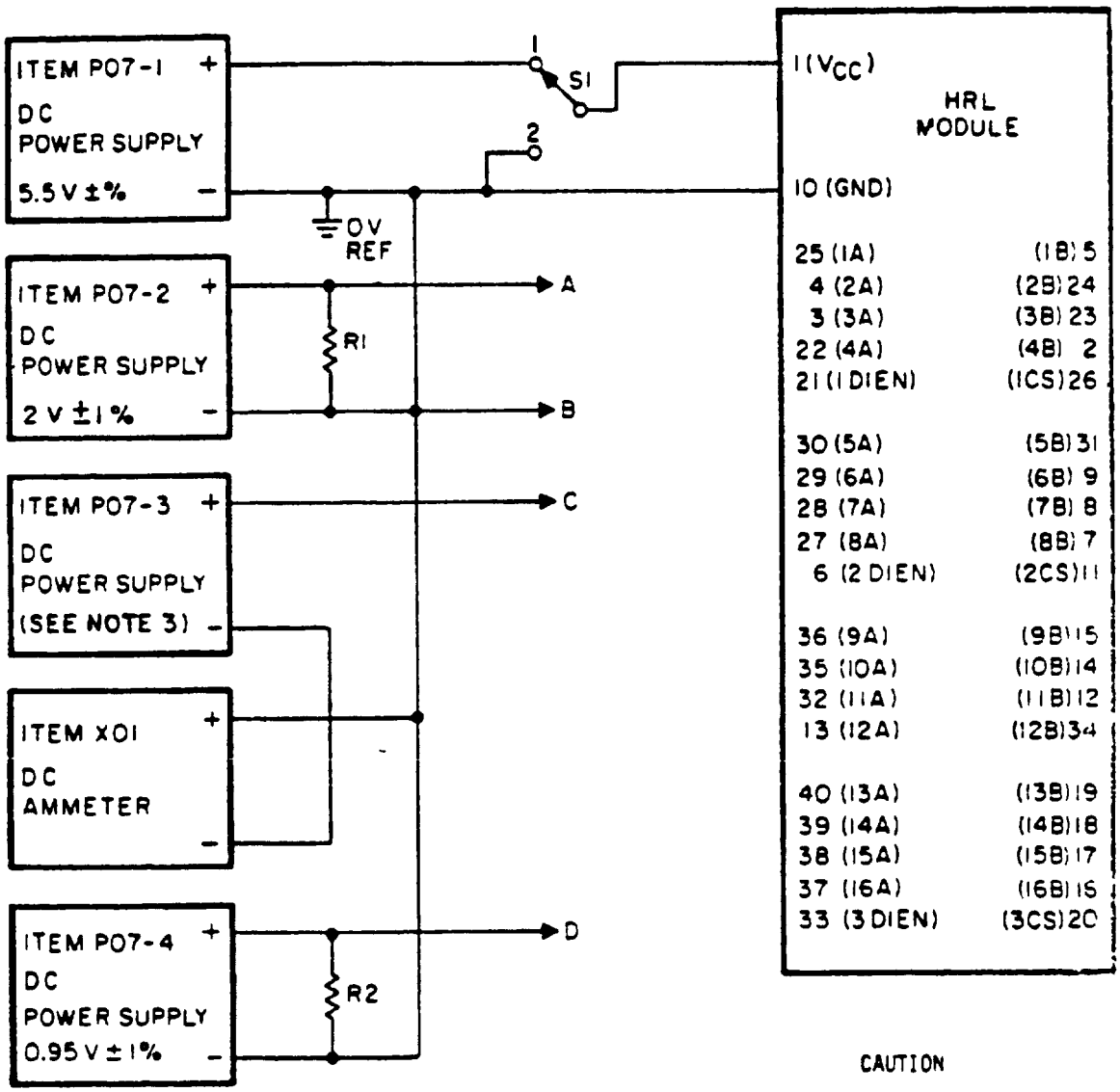
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E83	3A Output leakage current, high level	21,26	23	3	Open	0	60	uA	9
E84	4A "	21,26	2	22	"	"	"	"	"
E85	5A "	6,11	31	30	"	"	"	"	"
E86	6A "	"	9	29	"	"	"	"	"
E87	7A "	"	8	28	"	"	"	"	"
E88	8A "	"	7	27	"	"	"	"	"
E89	9A "	20,33	15	36	"	"	"	"	"
E90	10A "	"	14	35	"	"	"	"	"
E91	11A "	"	12	32	"	"	"	"	"
E92	12A "	"	34	13	"	"	"	"	"
E93	13A "	"	19	40	"	"	"	"	"
E94	14A "	"	18	39	"	"	"	"	"
E95	15A "	"	17	38	"	"	"	"	"
E96	16A "	"	16	37	"	"	"	"	"

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TABLE VIII. Output leakage current (power-off) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1A Output leakage current, power off	Open	Open	25	Open	0	60	µA	9
F2	2A "	"	"	4	"	"	"	"	"
F3	3A "	"	"	3	"	"	"	"	"
F4	4A "	"	"	22	"	"	"	"	"
F5	5A "	"	"	30	"	"	"	"	"
F6	6A "	"	"	29	"	"	"	"	"
F7	7A "	"	"	28	"	"	"	"	"
F8	8A "	"	"	27	"	"	"	"	"
F9	9A "	"	"	36	"	"	"	"	"
F10	10A "	"	"	35	"	"	"	"	"
F11	11A "	"	"	32	"	"	"	"	"
F12	12A "	"	"	13	"	"	"	"	"
F13	13A "	"	"	40	"	"	"	"	"
F14	14A "	"	"	39	"	"	"	"	"
F15	15A "	"	"	38	"	"	"	"	"
F16	16A "	"	"	37	"	"	"	"	"

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NOTES.
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. S1 = SPDT switch.
3. For test conditions E1-E48 and F1-F16 set item P07-3 to 2.40 V ±1% and for test conditions E49-E96 set item P07-3 to 3.4 V ±1%.
4. For test conditions E1-E96, switch S1 is in position 1 and for test conditions F1-F16, switch S1 is in position 2.

FIGURE 9. Output leakage current (high-level and power-off) test circuit for tables VII and VIII

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TABLE IX. Output leakage current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1B Output leakage current, low level	25	21,26	Open	5	-250	100	µA	10
G2	2B "	4	"	"	24	"	"	"	"
G3	3B "	3	"	"	23	"	"	"	"
G4	4B "	22	"	"	2	"	"	"	"
G5	5B "	30	6,11	"	31	"	"	"	"
G6	6B "	29	"	"	9	"	"	"	"
G7	7B "	28	"	"	8	"	"	"	"
G8	8B "	27	"	"	7	"	"	"	"
G9	9B "	36	33,20	"	15	"	"	"	"
G10	10B "	35	"	"	14	"	"	"	"
G11	11B "	32	"	"	12	"	"	"	"
G12	12B "	13	"	"	34	"	"	"	"
G13	13B "	40	"	"	19	"	"	"	"
G14	14B "	39	"	"	18	"	"	"	"
G15	15B "	38	"	"	17	"	"	"	"
G16	16B "	37	"	"	16	"	"	"	"
G17	1B "	Open	21	26	5	"	"	"	"
G18	2B "	"	"	"	24	"	"	"	"
G19	3B "	"	"	"	23	"	"	"	"
G20	4B "	"	"	"	2	"	"	"	"

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TABLE IX. Output leakage current (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G21	5B Output leakage current, low level	Open	6	11	31	-250	100	µA	10
G22	6B "	"	"	"	9	"	"	"	"
G23	7B "	"	"	"	8	"	"	"	"
G24	8B "	"	"	"	7	"	"	"	"
G25	9B "	"	33	20	15	"	"	"	"
G26	10B "	"	"	"	14	"	"	"	"
G27	11B "	"	"	"	12	"	"	"	"
G28	12B "	"	"	"	34	"	"	"	"
G29	13B "	"	"	"	19	"	"	"	"
G30	14B "	"	"	"	18	"	"	"	"
G31	15B "	"	"	"	17	"	"	"	"
G32	16B "	"	"	"	16	"	"	"	"
G33	1B "	25	26	21	5	"	"	"	"
G34	2B "	4	"	"	24	"	"	"	"
G35	3B "	3	"	"	23	"	"	"	"
G36	4B "	22	"	"	2	"	"	"	"
G37	5B "	30	11	6	31	"	"	"	"
G38	6B "	29	"	"	9	"	"	"	"
G39	7B "	28	"	"	8	"	"	"	"
G40	8B "	27	"	"	7	"	"	"	"

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TABLE IX. Output leakage current (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G41	9B Output leakage current, low level	36	20	33	15	-250	100	µA	10
G42	10B "	35	"	"	14	"	"	"	"
G43	11B "	32	"	"	12	"	"	"	"
G44	12B "	13	"	"	34	"	"	"	"
G45	13B "	40	"	"	19	"	"	"	"
G46	14B "	39	"	"	18	"	"	"	"
G47	15B "	38	"	"	17	"	"	"	"
G48	16B "	37	"	"	16	"	"	"	"
G49	1A "	5	21,26	Open	25	"	20	"	"
G50	2A "	24	"	"	4	"	"	"	"
G51	3A "	23	"	"	3	"	"	"	"
G52	4A "	2	"	"	22	"	"	"	"
G53	5A "	31	6,11	"	30	"	"	"	"
G54	6A "	9	"	"	29	"	"	"	"
G55	7A "	8	"	"	28	"	"	"	"
G56	8A "	7	"	"	27	"	"	"	"
G57	9A "	15	33,20	"	36	"	"	"	"
G58	10A "	14	"	"	35	"	"	"	"
G59	11A "	12	"	"	32	"	"	"	"
G60	12A "	34	"	"	13	"	"	"	"

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TABLE IX. Output leakage current (low-level) test - Continued.

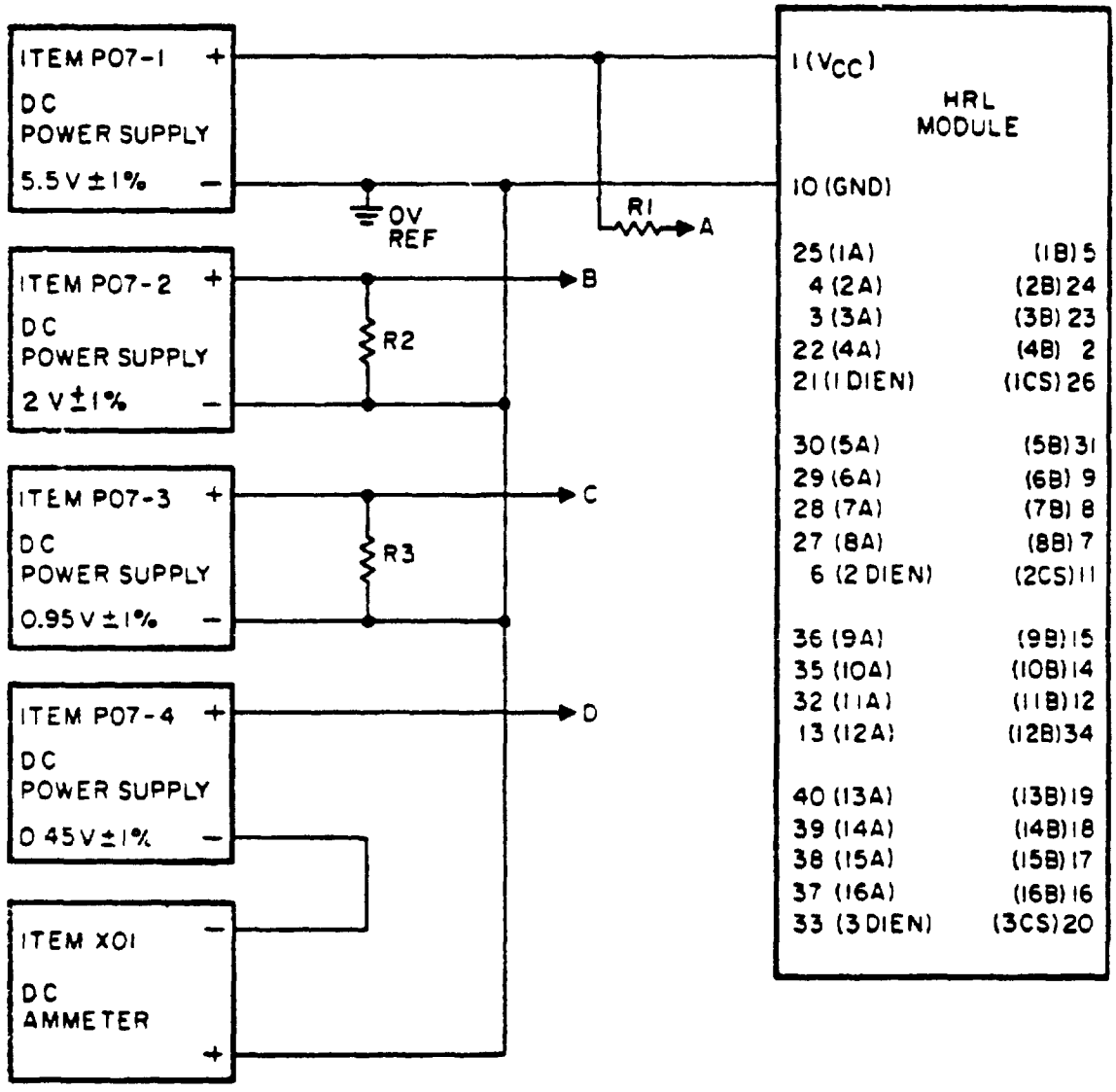
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G61	13A Output leakage current, low level	19	33,20	Open	40	-250	20	µA	10
G62	14A "	18	"	"	39	"	"	"	"
G63	15A "	17	"	"	38	"	"	"	"
G64	16A "	16	"	"	37	"	"	"	"
G65	1A "	5	26	21	25	"	"	"	"
G66	2A "	24	"	"		"	"	"	"
G67	3A "	23	"	"	3	"	"	"	"
G68	4A "	2	"	"	22	"	"	"	"
G69	5A "	31	11	6	30	"	"	"	"
G70	6A "	9	"	"	29	"	"	"	"
G71	7A "	8	"	"	28	"	"	"	"
G72	8A "	7	"	"	27	"	"	"	"
G73	9A "	15	20	33	36	"	"	"	"
G74	10A "	14	"	"	35	"	"	"	"
G75	11A "	12	"	"	32	"	"	"	"
G76	12A "	34	"	"	13	"	"	"	"
G77	13A "	19	"	"	40	"	"	"	"
G78	14A "	18	"	"	39	"	"	"	"
G79	15A "	17	"	"	38	"	"	"	"
G80	16A "	16	"	"	37	"	"	"	"

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TABLE IX. Output leakage current (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G81	1A Output leakage current, low level	Open	Open	21,26	25	-250	20	µA	10
G82	2A "	"	"	"	4	"	"	"	"
G83	3A "	"	"	"	3	"	"	"	"
G84	4A "	"	"	"	22	"	"	"	"
G85	5A "	"	"	6,11	30	"	"	"	"
G86	6A "	"	"	"	29	"	"	"	"
G87	7A "	"	"	"	28	"	"	"	"
G88	8A "	"	"	"	27	"	"	"	"
G89	9A "	"	"	33,20	36	"	"	"	"
G90	10A "	"	"	"	35	"	"	"	"
G91	11A "	"	"	"	32	"	"	"	"
G92	12A "	"	"	"	13	"	"	"	"
G93	13A "	"	"	"	40	"	"	"	"
G94	14A "	"	"	"	39	"	"	"	"
G95	15A "	"	"	"	38	"	"	"	"
G96	16A "	"	"	"	37	"	"	"	"

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NOTES
1. R1 = 4.99 kΩ ±1%, 1/8 W
2. R2 = R3 = 51.1Ω ±5%, 1/2 W.

CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground

FIGURE 10 Output leakage current (low-level) test circuit for table IX

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TABLE X. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1B Output voltage, low level (I_{OL} = 25 mA)	4,3,22	21,26,25	5	0.00	0.45	V	11,14
H2	2B "	25,3,22	21,26,4	24	"	"	"	"
H3	3B "	25,4,22	21,26,3	23	"	"	"	"
H4	4B "	25,4,3	21,26,22	2	"	"	"	"
H5	5B "	29,28,27	6,11,30	31	"	"	"	"
H6	6B "	30,28,27	6,11,29	9	"	"	"	"
H7	7B "	30,29,27	6,11,28	8	"	"	"	"
H8	8B "	30,29,28	6,11,27	7	"	"	"	"
H9	9B "	35,32,13, 40,39,38, 37	33,20,36	15	"	"	"	"
H10	10B "	36,32,13, 40,39,38, 37	33,20,35	14	"	"	"	"
H11	11B "	36,35,13, 40,39,38, 37	33,20,32	12	"	"	"	"
H12	12B "	36,35,32, 40,39,38, 37	33,20,13	34	"	"	"	"
H13	13B "	36,35,32, 13,39,38, 37	33,20,40	19	"	"	"	"
H14	14B "	36,35,32, 13,40,38, 37	33,20,39	18	"	"	"	"

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TABLE X. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H15	15B Output voltage, low level (I_{OL} = 25 mA)	36,35,32, 13,40,39, 37	33,20,38	17	0.00	0.45	V	11,14
H16	16B Output voltage, low level (I_{OL} = 25 mA)	36,35,32, 13,40,39, 38	33,20,37	16	"	"	"	11,14
H17	1A Output voltage, low level (I_{OL} = 15 mA)	21,24,23,2	26,5	25	"	"	"	11,13
H18	2A "	21,5,23,2	26,24	4	"	"	"	"
H19	3A "	21,5,24,2	26,23	3	"	"	"	"
H20	4A "	21,5,24,23	26,2	22	"	"	"	"
H21	5A "	6,9,8,7	11,31	30	"	"	"	"
H22	6A "	6,31,8,7	11,9	29	"	"	"	"
H23	7A "	6,31,9,7	11,8	28	"	"	"	"
H24	8A "	6,31,9,8	11,7	27	"	"	"	"
H25	9A "	33,14,12, 34,19,18, 17,16	20,15	36	"	"	"	"
H26	10A "	33,15,12, 34,19,18, 17,16	20,14	35	"	"	"	"
H27	11A "	33,15,14, 34,19,18, 17,16	20,12	32	"	"	"	"
H28	12A "	33,15,14, 34,19,18, 17,16	20,34	13	"	"	"	"

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TABLE X. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H29	13A Output voltage, low level (I_{OL} = 15 mA)	33,15,14, 12,34,18, 17,16	20,19	40	0.00	0.45	V	11,13
H30	14A "	33,15,14, 12,34,19, 17,16	20,18	39	"	"	"	"
H31	15A "	33,15,14, 12,34,19, 18,16	20,17	38	"	"	"	"
H32	16A "	33,15,14, 12,34,19, 18,17	20,16	37	"	"	"	"
H33	1B Output voltage, low level (I_{OL} = 45 mA)	4,3,22	21,26,25	5	"	0.60	"	12
H34	2B "	25,3,22	21,26,4	24	"	"	"	"
H35	3B "	25,4,22	21,26,3	23	"	"	"	"
H36	4B "	25,4,3	21,26,22	2	"	"	"	"
H37	5B "	29,28,27	6,11,30	31	"	"	"	"
H38	6B "	30,28,27	6,11,29	9	"	"	"	"
H39	7B "	30,29,27	6,11,28	8	"	"	"	"
H40	8B "	30,29,28	6,11,27	7	"	"	"	"
H41	9B "	35,32,13 40,39,38, 37	33,20,36	15	"	"	"	"
H42	10B "	36,32,13 40,39,38, 37	33,20,35	14	"	"	"	"

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TABLE X. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H43	11B Output voltage, low level (I_{OL} = 45 mA)	36,35,13, 40,39,38, 37	33,20,32	12	0.00	0.60	V	12
H44	12B "	36,35,32, 40,39,38, 37	33,20,13	34	"	"	"	"
H45	13B "	36,35,32, 13,39,38, 37	33,20,40	19	"	"	"	"
H46	14B "	36,35,32, 13,40,38, 37	33,20,39	18	"	"	"	"
H47	15B "	36,35,32, 13,40,39, 37	33,20,38	17	"	"	"	"
H48	16B "	36,35,32, 13,40,39, 38	33,20,37	16	"	"	"	"

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TABLE XI. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1A Output voltage, high level (I_{OH} = -0.5 mA)	21,5	26,24,23,2	25	3.4	4.5	V	11,13
J2	2A "	21,24	26,5,23,2	4	"	"	"	"
J3	3A "	21,23	26,5,24,2	3	"	"	"	"
J4	4A "	21,2	26,5,24,23	22	"	"	"	"
J5	5A "	6,31	11,9,8,7	30	"	"	"	"
J6	6A "	6,9	11,31,8,7	29	"	"	"	"
J7	7A "	6,8	11,31,9,7	28	"	"	"	"
J8	8A "	6,7	11,31,9,8	27	"	"	"	"
J9	9A "	33,15	20,14,12, 34,19,18, 17,16	36	"	"	"	"
J10	10A "	33,14	20,15,12, 34,19,18, 17,16	35	"	"	"	"
J11	11A "	33,12	20,15,14, 34,19,18, 17,16	32	"	"	"	"
J12	12A "	33,34	20,15,14, 12,19,18, 17,16	13	"	"	"	"
J13	13A "	33,19	20,15,14, 12,34,18, 17,16	40	"	"	"	"
J14	14A "	33,18	20,15,14, 12,34,19, 17,16	39	"	"	"	"

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TABLE XI. Output voltage (high-level) test - Continued.

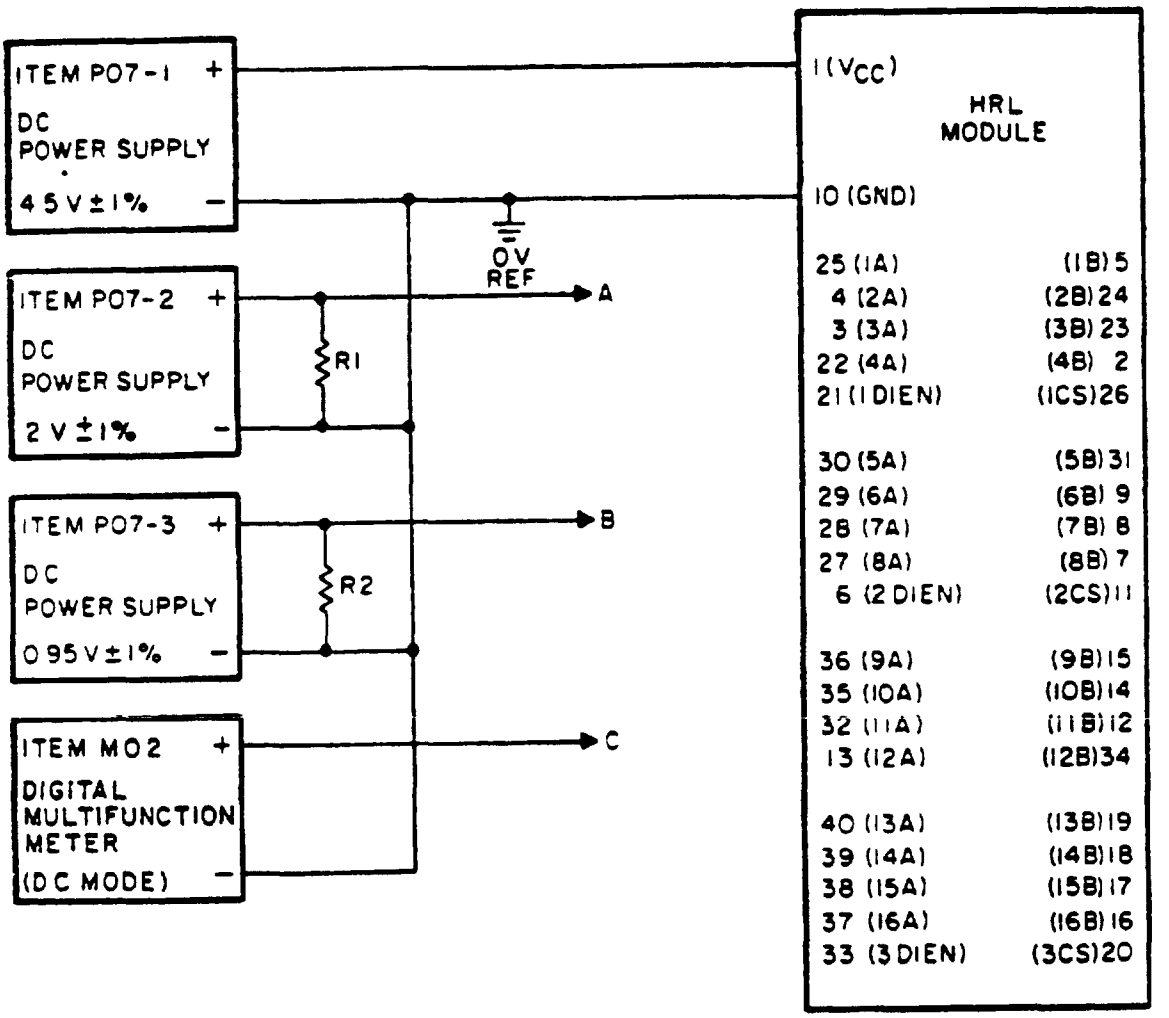
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J15	15A Output voltage, high level (I_{OH} = -0.5 mA)	33,17	20,15,14, 12,34,19, 18,16	38	3.4	4.5	V	11,13
J16	16A Output voltage, high level (I_{OH} = -0.5 mA)	33,16	20,15,14, 12,34,19, 18,17	37	3.4	"	"	11,13
J17	1B Output voltage, high level (I_{OH} = -5 mA)	25	21,26,4,3 22	5	2.4	"	"	11,14
J18	2B "	4	21,26,25, 3,22	24	"	"	"	"
J19	3B "	3	21,26,25, 4,22	23	"	"	"	"
J20	4B "	22	21,26,25, 4,3	2	"	"	"	"
J21	5B "	30	6,11,29,28 27	31	"	"	"	"
J22	6B "	29	6,11,30,28 27	9	"	"	"	"
J23	7B "	28	6,11,30,29 27	8	"	"	"	"
J24	8B "	27	6,11,30,29 28	7	"	"	"	"
J25	9B "	36	20,33,35, 32,13,40, 39,38,37	15	"	"	"	"
J26	10B "	35	20,33,36, 32,13,40, 39,38,37	14	"	"	"	"
J27	11B "	12	20,33,36, 35,13,40, 39,38,37	12	"	"	"	"

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TABLE XI. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J28	12B Output voltage, high level (I_{OH} = -5 mA)	13	20,33,36, 35,32,40, 39,38,37	34	2.4	4.5	V	11,14
J29	13B "	40	20,33,36, 35,32,13, 39,38,37	19	"	"	"	"
J30	14B "	39	20,33,36, 35,32,13, 40,38,37	18	"	"	"	"
J31	15B "	38	20,33,36, 35,32,13, 40,39,37	17	"	"	"	"
J32	16B "	37	20,33,36, 35,32,13, 40,39,38	16	"	"	"	"

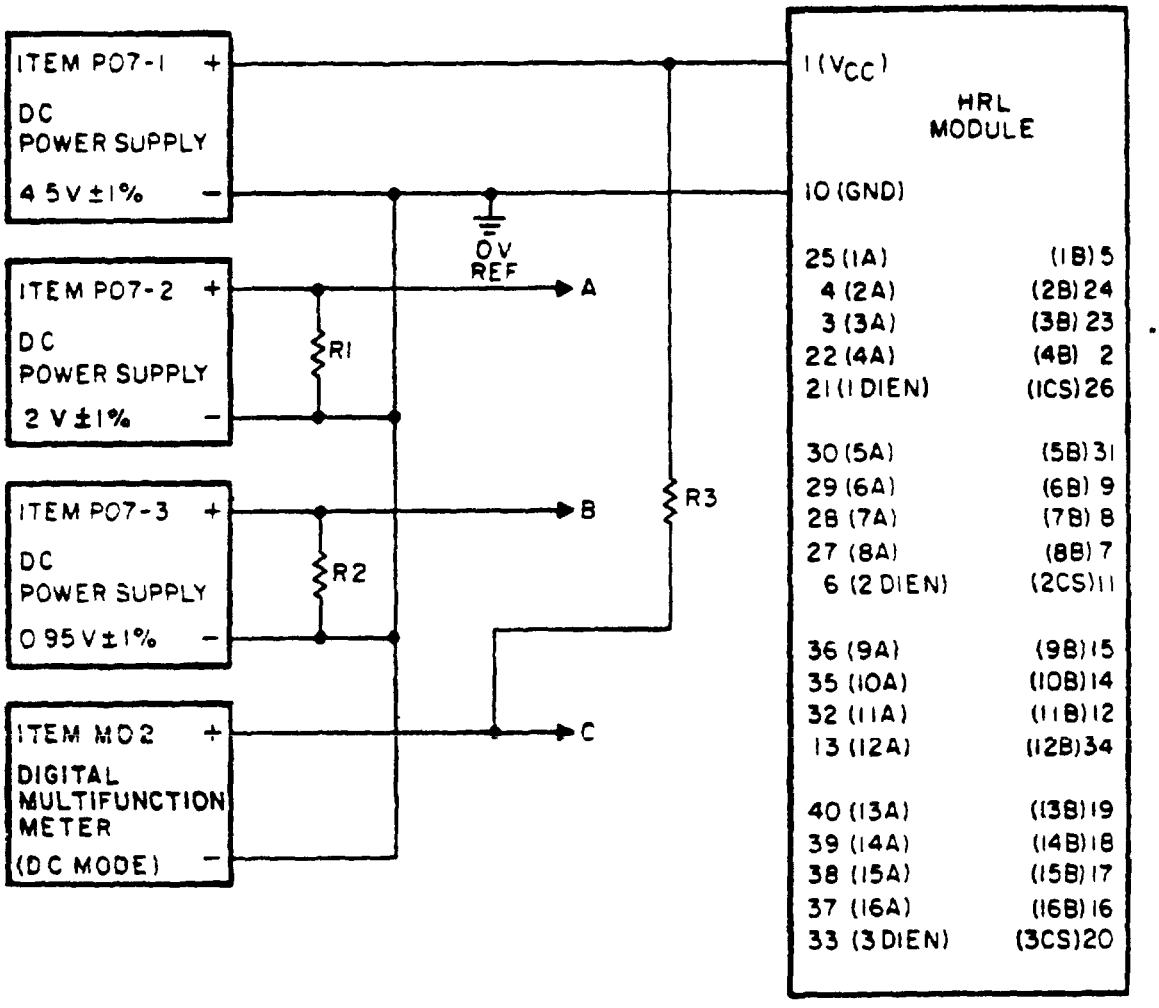
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NOTES:
1. R1=R2 = 51.1Ω ±5%, 1/2 W.
2. Load output under test with load circuit as specified in test figure column of tables X and XI.

FIGURE 11. Output voltage (low-level and high-level) test circuit for tables X and XI.

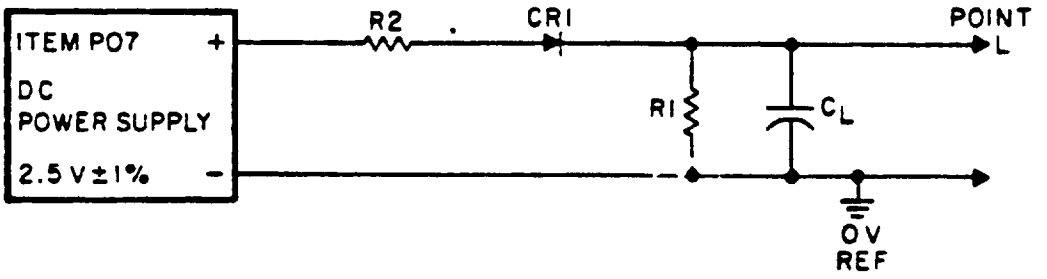
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NOTES
1. R1=R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 86 6Ω ±1%, 1/4 W

* FIGURE 12. Output voltage (low-level) test circuit for table X.

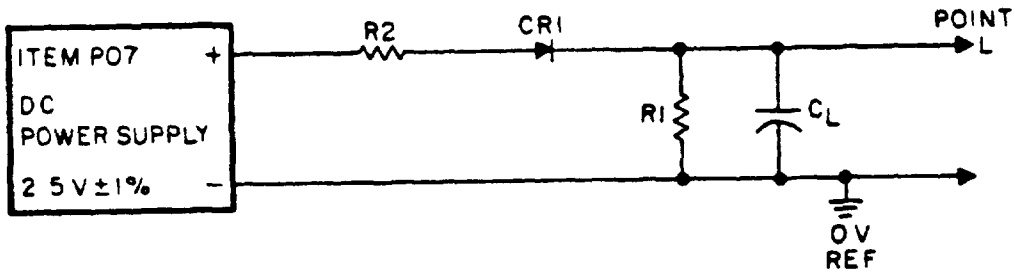
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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 6.81 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 15 mA ±5% out of point L with point L held at 0.4 V
4. C_L = 50 ±5 pF (including probe and parasitic capacitance).
5. Capacitor C_L is not required for dc electrical test conditions H17-H32 and J1-J16.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm

FIGURE 13. HRL logic load circuit for "A" outputs.

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NOTES
1. CR1 = 1N4150, or equivalent.
2. R1 = 475Ω ±1 , 1/8 w.
3. R2 = Select value for a current flow of 25 mA ±5. out of point L with point L held at 0.4[illegible] V.
4. For test conditions N1-N16, P1-P16, Q1-Q16, R1-R16, S1-S16, T1-T16, U1-U16, V1-V16, W1-W16, and X1-X16, C_L = 50 ±5 pF (including probe and parasitic capacitance). For test conditions Q33-Q48 and R33-R48, C_L = 1,000 ±100 pF (including probe and parasitic capacitance).
5. Capacitor C_L is not required for dc electrical test conditions H1-H16 and J17-J32.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 m .

* FIGURE 14. HRL logic load circuit for "B" outputs.

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TABLE XII. Input current (high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
K1	1DIEN Input current, high level	21	0	80	µA	15
K2	1CS "	26	"	"	"	"
K3	2DIEN "	6	"	"	"	"
K4	2CS "	11	"	"	"	"
K5	3DIEN "	33	"	160	"	"
K6	3CS "	20	"	160	"	"

TABLE XIII. Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
L1	1DIEN Input current, low level	21	-10	-550	µA	16
L2	1CS "	26	"	"	"	"
L3	2DIEN "	6	"	"	"	"
L4	2CS "	11	"	"	"	"
L5	3DIEN "	33	"	-1,100	"	"
L6	3CS "	20	"	-1,100	"	"

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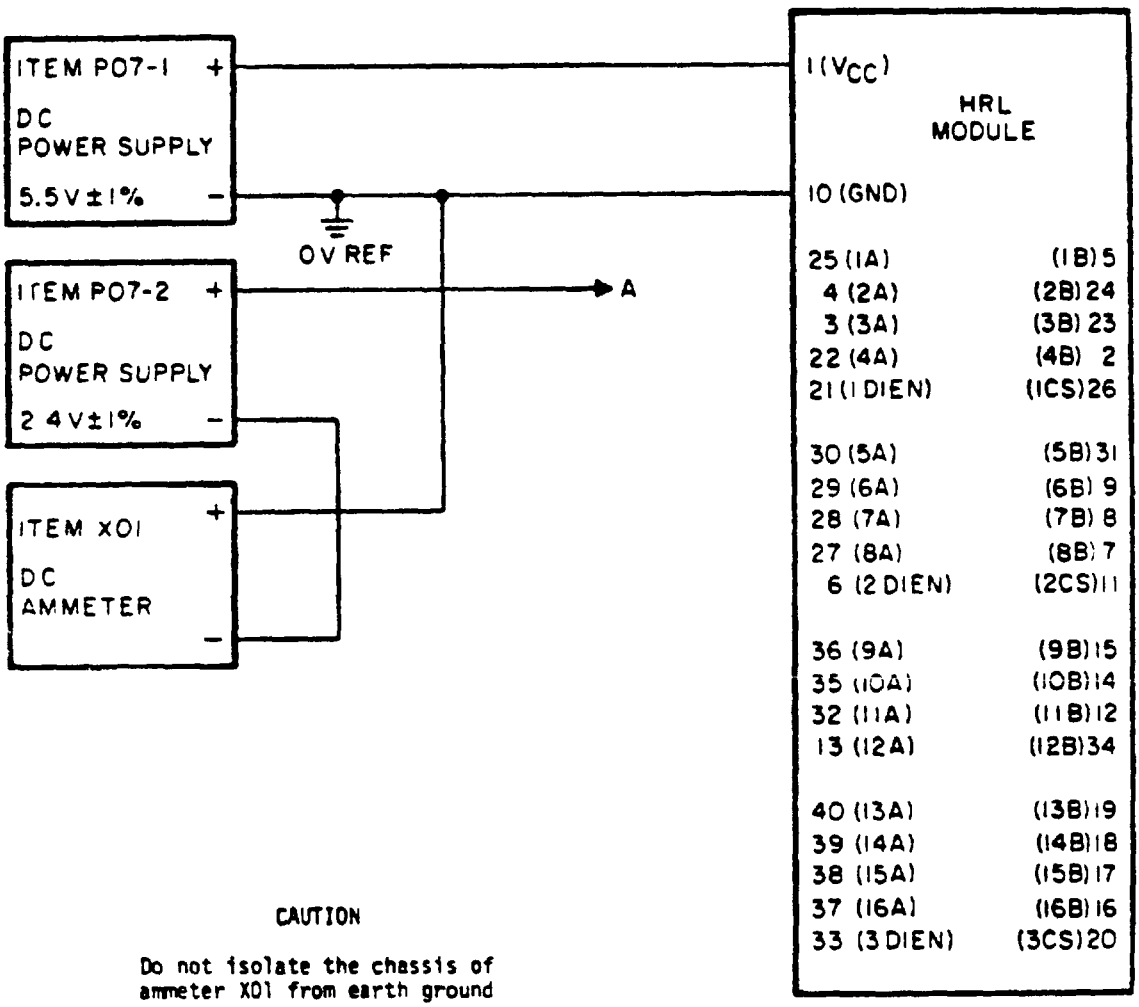


FIGURE 15. Input current (high-level) test circuit for table XII

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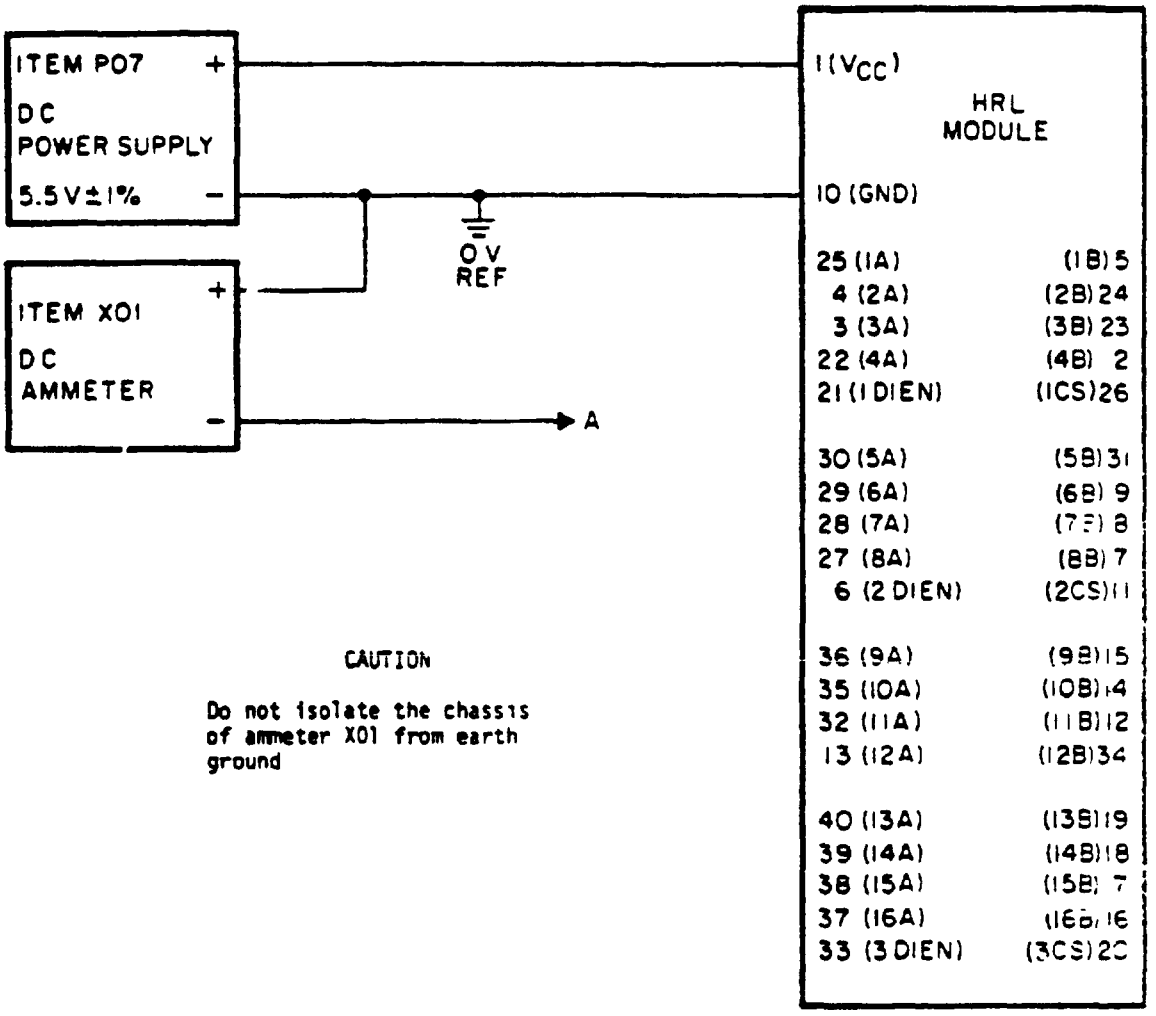


FIGURE 16. Input current (low-level) test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current	50	520	mA	17

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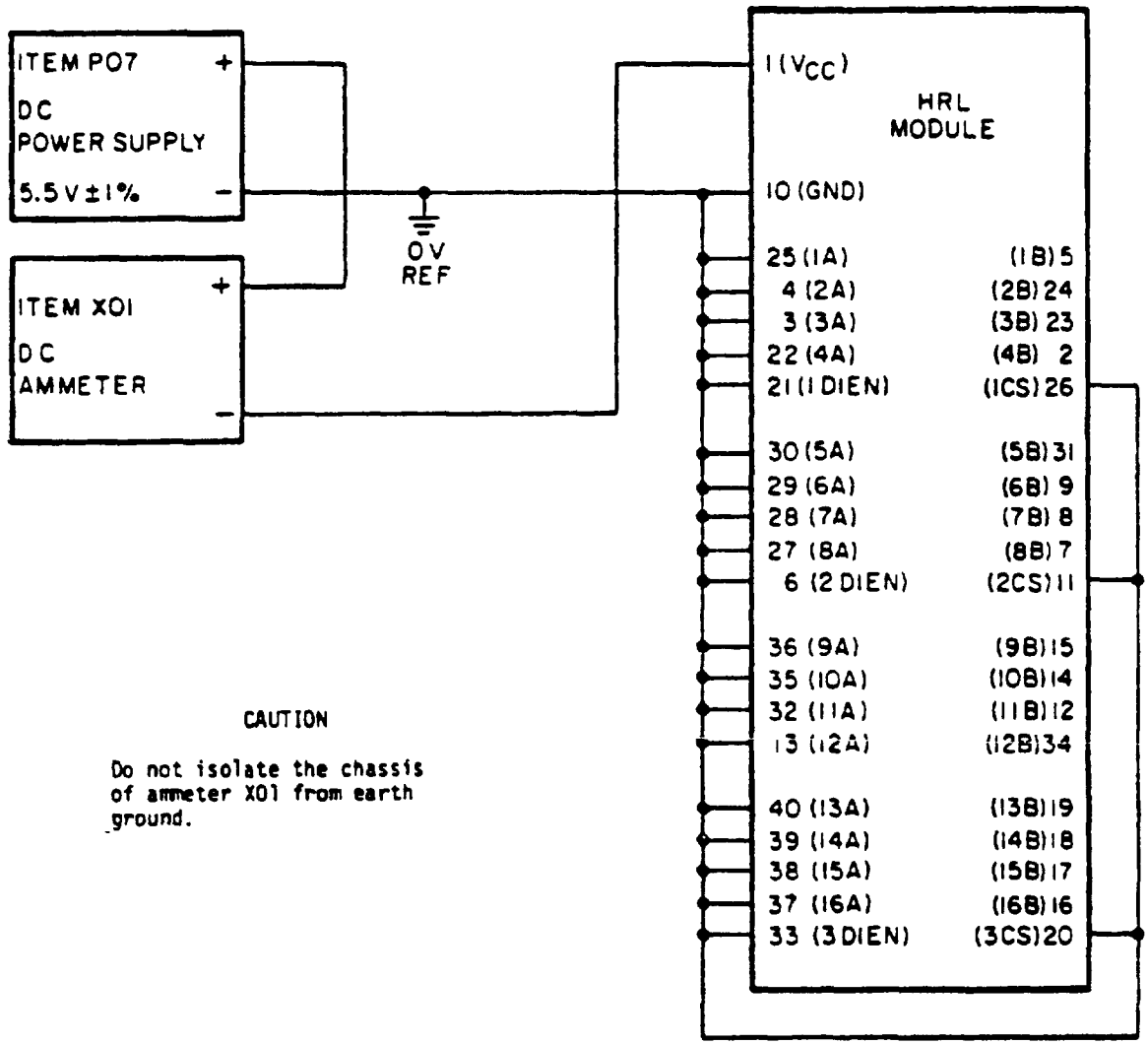


FIGURE 17. Power supply current test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH}) test

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1B Transition time (t_{TLH}), C_L = 50 pF	25	21	5	Open	2	25	ns	14,18,19
N2	2B "	4	"	24	"	"	"	"	"
N3	3B "	3	"	23	"	"	"	"	"
N4	4B "	22	"	2	"	"	"	"	"
N5	5B "	30	6	31	"	"	"	"	"
N6	6B "	29	"	9	"	"	"	"	"
N7	7B "	28	"	8	"	"	"	"	"
N8	8B "	27	"	7	"	"	"	"	"
N9	9B "	36	33	15	"	"	"	"	"
N10	10B "	35	"	14	"	"	"	"	"
N11	11B "	32	"	12	"	"	"	"	"
N12	12B "	13	"	34	"	"	"	"	"
N13	13B "	40	"	19	"	"	"	"	"
N14	14B "	39	"	18	"	"	"	"	"
N15	15B "	38	"	17	"	"	"	"	"
N16	16B "	37	"	16	"	"	"	"	"
N17	1A "	5	Open	25	21	5	35	"	13,18,19
N18	2A "	24	"	4	"	"	"	"	"
N19	3A "	23	"	3	"	"	"	"	"
N20	4A "	2	"	22	"	"	"	"	"
N21	5A "	31	"	30	6	"	"	"	"

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TABLE XV. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N22	6A Transition time (t_{TLH}), C_L = 50 pF	9	Open	29	6	5	35	ns	13,18,19
N23	7A "	8	"	28	"	"	"	"	"
N24	8A "	7	"	27	"	"	"	"	"
N25	9A "	15	"	36	33	"	"	"	"
N26	10A "	14	"	35	"	"	"	"	"
N27	11A "	12	"	32	"	"	"	"	"
N28	12A "	34	"	13	"	"	"	"	"
N29	13A "	19	"	40	"	"	"	"	"
N30	14A "	18	"	39	"	"	"	"	"
N31	15A "	17	"	38	"	"	"	"	"
N32	16A "	16	"	37	"	"	"	"	"

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TABLE XVI. Transition time (t_{THL}) test

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1B Transition time (t_{THL}), C_L = 50 pF	25	21	5	Open	2	10	ns	14,18,19
P2	2B "	4	"	24	"	"	"	"	"
P3	3B "	3	"	23	"	"	"	"	"
P4	4B "	22	"	2	"	"	"	"	"
P5	5B "	30	6	31	"	"	"	"	"
P6	6B "	29	"	9	"	"	"	"	"
P7	7B "	28	"	8	"	"	"	"	"
P8	8B "	27	"	7	"	"	"	"	"
P9	9B "	36	33	15	"	"	"	"	"
P10	10B "	35	"	14	"	"	"	"	"
P11	11B "	32	"	12	"	"	"	"	"
P12	12B "	13	"	34	"	"	"	"	"
P13	13B "	40	"	19	"	"	"	"	"
P14	14B "	39	"	18	"	"	"	"	"
P15	15B "	38	"	17	"	"	"	"	"
P16	16B "	37	"	16	"	"	"	"	"
P17	1A "	5	Open	25	21	"	15	"	13,18,19
P18	2A "	24	"	4	"	"	"	"	"
P19	3A "	23	"	3	"	"	"	"	"
P20	4A "	2	"	22	"	"	"	"	"

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TABLE XVI. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	" to pin	C to pin	D to pin	Min	Max		
P21	5A Transition time (t_{THL}), C_L = 50 pF	31	Open	30	6	2	15	ns	13,18,19
P22	6A "	9	"	29	"	"	"	"	"
P23	7A "	8	"	28	"	"	"	"	"
P24	8A "	7	"	27	"	"	"	"	"
P25	9A "	15	"	36	33	"	"	"	"
P26	10A "	14	"	35	"	"	"	"	"
P27	11A "	12	"	32	"	"	"	"	"
P28	12A "	34	"	13	"	"	"	"	"
P29	13A "	19	"	40	"	"	"	"	"
P30	14A "	18	"	39	"	"	"	"	"
P31	15A "	17	"	38	"	"	"	"	"
P32	16A "	16	"	37	"	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1A to 1B Propagation delay time (t_{PLH}), C_L = 50 pF	25	21	5	Open	2	28	ns	14,18,19
Q2	2A to 2B "	4	"	24	"	"	"	"	"
Q3	3A to 3B "	3	"	23	"	"	"	"	"
Q4	4A to 4B "	22	"	2	"	"	"	"	"
Q5	5A to 5B "	30	6	31	"	"	"	"	"
Q6	6A to 6B "	29	"	9	"	"	"	"	"
Q7	7A to 7B "	28	"	8	"	"	"	"	"
Q8	8A to 8B "	27	"	7	"	"	"	"	"
Q9	9A to 9B "	36	33	15	"	"	"	"	"
Q10	10A to 10B "	35	"	14	"	"	"	"	"
Q11	11A to 11B "	32	"	12	"	"	"	"	"
Q12	12A to 12B "	13	"	34	"	"	"	"	"
Q13	13A to 13B "	40	"	19	"	"	"	"	"
Q14	14A to 14B "	39	"	18	"	"	"	"	"
Q15	15A to 15B "	38	"	17	"	"	"	"	"
Q16	16A to 16B "	37	"	16	"	"	"	"	"
Q17	1B to 1A "	5	Open	25	21	"	"	"	13,18,19
Q18	2B to 2A "	24	"	4	"	"	"	"	"
Q19	3B to 3A "	23	"	3	"	"	"	"	"
Q20	4B to 4A "	2	"	22	"	"	"	"	"
Q21	5B to 5A "	31	"	30	6	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q22	6B to 6A Propagation delay time (t_{PLH}), C_L = 50 pF	9	Open	29	6	2	28	ns	13,18,19
Q23	7B to 7A "	8	"	28	"	"	"	"	"
Q24	8B to 8A "	7	"	27	"	"	"	"	"
Q25	9B to 9A "	15	"	36	33	"	"	"	"
Q26	10B to 10A "	14	"	35	"	"	"	"	"
Q27	11B to 11A "	12	"	32	"	"	"	"	"
Q28	12B to 12A "	34	"	13	"	"	"	"	"
Q29	13B to 13A "	19	"	40	"	"	"	"	"
Q30	14B to 14A "	18	"	39	"	"	"	"	"
Q31	15B to 15A "	17	"	38	"	"	"	"	"
Q32	16B to 16A "	16	"	37	"	"	"	"	"
Q33	1A to 1B Propagation delay time (t_{PLH}), C_L = 1,000 pF	25	21	5	Open	10	100	"	14,18,19
Q34	2A to 2B "	4	"	24	"	"	"	"	"
Q35	3A to 3B "	3	"	23	"	"	"	"	"
Q36	4A to 4B "	22	"	2	"	"	"	"	"
Q37	5A to 5B "	30	6	31	"	"	"	"	"
Q38	6A to 6B "	29	"	9	"	"	"	"	"
Q39	7A to 7B "	28	"	8	"	"	"	"	"
Q40	8A to 8B "	27	"	7	"	"	"	"	"
Q41	9A to 9B "	36	33	15	"	"	"	"	"

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TABLE XVII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q42	10A to 10B Propagation delay time (t_{PLH}), C_L = 1,000 pF	35	33	14	Open	10	100	ns	14,18,19
Q43	11A to 11B "	32	"	12	"	"	"	"	"
Q44	12A to 12B "	13	"	34	"	"	"	"	"
Q45	13A to 13B "	40	"	19	"	"	"	"	"
Q46	14A to 14B "	39	"	18	"	"	"	"	"
Q47	15A to 15B "	38	"	17	"	"	"	"	"
Q48	16A to 16B "	37	"	16	"	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1A to 1B Propagation delay time (t_{PHL}), C_L = 50 pF	25	21	5	Open	2	28	ns	14,18,19
R2	2A to 2B "	4	"	24	"	"	"	"	"
R3	3A to 3B "	3	"	23	"	"	"	"	"
R4	4A to 4B "	22	"	2	"	"	"	"	"
R5	5A to 5B "	30	6	31	"	"	"	"	"
R6	6A to 6B "	29	"	9	"	"	"	"	"
R7	7A to 7B "	28	"	8	"	"	"	"	"
R8	8A to 8B "	27	"	7	"	"	"	"	"
R9	9A to 9B "	36	33	15	"	"	"	"	"
R10	10A to 10B "	35	"	14	"	"	"	"	"
R11	11A to 11B "	32	"	12	"	"	"	"	"
R12	12A to 12B "	13	"	34	"	"	"	"	"
R13	13A to 13B "	40	"	19	"	"	"	"	"
R14	14A to 14B "	39	"	18	"	"	"	"	"
R15	15A to 15B "	38	"	17	"	"	"	"	"
R16	16A to 16B "	37	"	16	"	"	"	"	"
R17	1B to 1A "	5	Open	25	21	"	"	"	13,18,19
R18	2B to 2A "	24	"	4	"	"	"	"	"
R19	3B to 3A "	23	"	3	"	"	"	"	"
R20	4B to 4A "	2	"	22	"	"	"	"	"
R21	5B to 5A "	31	"	30	6	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

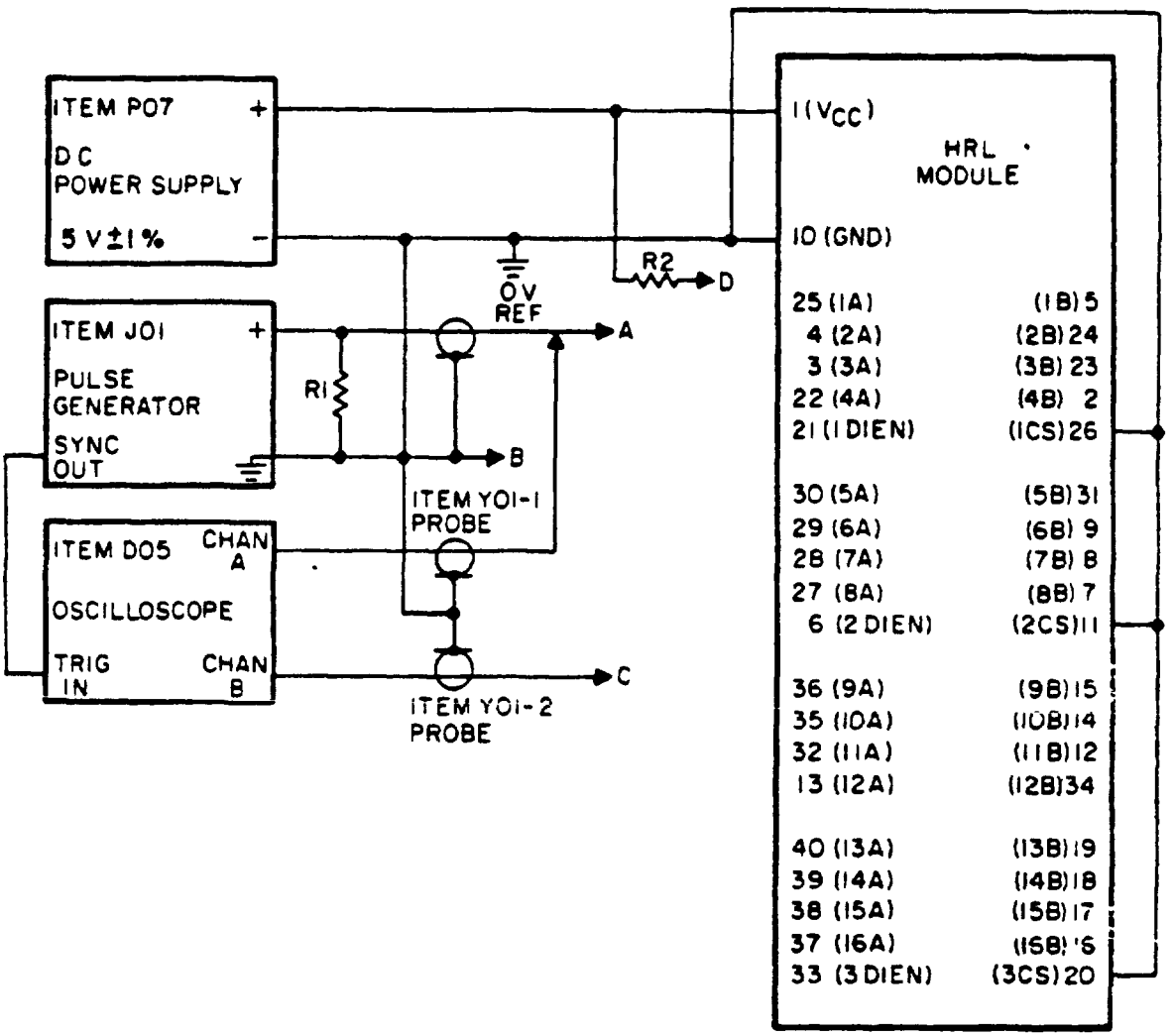
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R22	6B to 6A Propagation delay time (t_{PHL}), C_L = 50 pF	9	Open	29	6	2	28	ns	13,18,19
R23	7B to 7A "	8	"	28	"	"	"	"	"
R24	8B to 8A "	7	"	27	"	"	"	"	"
R25	9B to 9A "	15	"	36	33	"	"	"	"
R26	10B to 10A "	14	"	35	"	"	"	"	"
R27	11B to 11A "	12	"	32	"	"	"	"	"
R28	12B to 12A "	34	"	13	"	"	"	"	"
R29	13B to 13A "	19	"	40	"	"	"	"	"
R30	14B to 14A "	18	"	39	"	"	"	"	"
R31	15B to 15A "	17	"	38	"	"	"	"	"
R32	16B to 16A "	16	"	37	"	"	"	"	"
R33	1A to 1B Propagation delay time (t_{PHL}), C_L = 1,000 pf	25	21	5	Open	10	100	"	14,18,19
R34	2A to 2B "	4	"	24	"	"	"	"	"
R35	3A to 3B "	3	"	23	"	"	"	"	"
R36	4A to 4B "	22	"	2	"	"	"	"	"
R37	5A to 5B "	30	6	31	"	"	"	"	"
R38	6A to 6B "	29	"	9	"	"	"	"	"
R39	7A to 7B "	28	"	8	"	"	"	"	"
R40	8A to 8B "	27	"	7	"	"	"	"	"
R41	9A to 9B "	36	33	15	"	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R42	10A to 10B	Propagation delay time (t_{PHL}), C_L = 1,000 pF	35	33	14	Open	10	100	ns	14,18,19
R43	11A to 11B	"	32	"	12	"	"	"	"	"
R44	12A to 12B	"	13	"	34	"	"	"	"	"
R45	13A to 13B	"	40	"	19	"	"	"	"	"
R46	14A to 14B	"	39	"	18	"	"	"	"	"
R47	15A to 15B	"	38	"	17	"	"	"	"	"
R48	16A to 16B	"	37	"	16	"	"	"	"	"

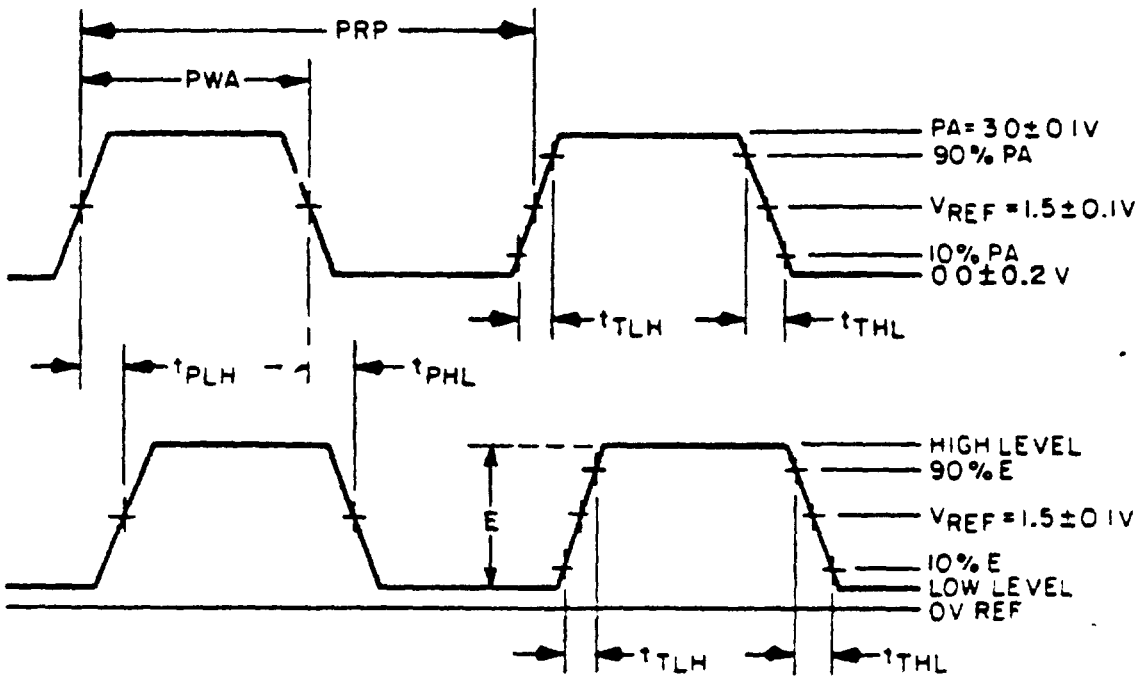
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NOTES.
1. R1 = 51 1Ω ±5%, 1/2 W.
2. R2 = 4.99 kΩ ±1%, 1/8 W.
3. Load output under test with load circuit as specified in test figure column of tables XV, XVI, XVII, and XVIII.

FIGURE 18 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV, XVI, XVII, and XVIII

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NOTES.

1. Input pulse parameters
   PWA = 500 ±50 ns.
   PRP = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL} = 10^{+9}$ ns.
2. Before making the measurements specified in tables XV, XVI, XVII, and XVIII verify that the output pulse high level is ≥ 3.4 V and the low level is ≤ 0.45 V for test conditions N17-N32, P17-P32, Q17-Q32, R17-R32 and that the high level is ≥ 2.4 V and the low level is ≤ 0.45 V for test conditions N1-N16, P1-P16, Q1-Q16, Q33-Q48, R1-R16, and R33-R48.
3. The lead length of electrical conductors between resistor R1 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 19. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XV, XVI, XVII, and XVIII

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TABLE XIX. Output disable delay time (t_{OH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1CS to 1B Output disable delay time (t_{OH}), C_L = 50 pF	26	21,25	5	Open	4	45	ns	14,20,21
S2	1CS to 2B "	"	21,4	24	"	"	"	"	"
S3	1CS to 3B "	"	21,3	23	"	"	"	"	"
S4	1CS to 4B "	"	21,22	2	"	"	"	"	"
S5	2CS to 5B "	11	6,30	31	"	"	"	"	"
S6	2CS to 6B "	"	6,29	9	"	"	"	"	"
S7	2CS to 7B "	"	6,28	8	"	"	"	"	"
S8	2CS to 8B "	"	6,27	7	"	"	"	"	"
S9	3CS to 9B "	20	33,36	15	"	"	"	"	"
S10	3CS to 10B "	"	33,35	14	"	"	"	"	"
S11	3CS to 11B "	"	33,32	12	"	"	"	"	"
S12	3CS to 12B "	"	33,13	34	"	"	"	"	"
S13	3CS to 13B "	"	33,40	19	"	"	"	"	"
S14	3CS to 14B "	"	33,39	18	"	"	"	"	"
S15	3CS to 15B "	"	33,38	17	"	"	"	"	"
S16	3CS to 16B "	"	33,37	16	"	"	"	"	"
S17	1CS to 1A "	26	5	25	21	"	"	"	13,20,21
S18	1CS to 2A "	"	24	4	"	"	"	"	"
S19	1CS to 3A "	"	23	3	"	"	"	"	"
S20	1CS to 4A "	"	2	22	"	"	"	"	"
S21	2CS to 5A "	11	31	30	6	"	"	"	"

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TABLE XIX. Output disable delay time (t_{OH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S22	2CS to 6A Output disable delay time (t_{OH}), C_L = 50 pF	11	9	29	6	4	45	ns	13,20,21
S23	3CS to 7A "	"	8	28	"	"	"	"	"
S24	3CS to 8A "	"	7	27	"	"	"	"	"
S25	3CS to 9A "	20	15	36	33	"	"	"	"
S26	3CS to 10A "	"	14	35	"	"	"	"	"
S27	3CS to 11A "	"	12	32	"	"	"	"	"
S28	3CS to 12A "	"	34	13	"	"	"	"	"
S29	3CS to 13A "	"	19	40	"	"	"	"	"
S30	3CS to 14A "	"	18	39	"	"	"	"	"
S31	3CS to 15A "	"	17	38	"	"	"	"	"
S32	3CS to 16A "	"	16	37	"	"	"	"	"

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TABLE XX. Output disable delay time (t_{HO}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1CS to 1B Output disable delay time (t_{HO}), C_L = 50 pF	26	21,25	5	Open	7	78	ns	14,20,21
T2	1CS to 2B "	"	21,4	24	"	"	"	"	"
T3	1CS to 3B "	"	21,3	23	"	"	"	"	"
T4	1CS to 4B "	"	21,22	2	"	"	"	"	"
T5	2CS to 5B "	11	6,30	31	"	"	"	"	"
T6	2CS to 6B "	"	6,29	9	"	"	"	"	"
T7	2CS to 7B "	"	6,28	8	"	"	"	"	"
T8	2CS to 8B "	"	6,27	7	"	"	"	"	"
T9	3CS to 9B "	20	33,36	15	"	"	"	"	"
T10	3CS to 10B "	"	33,35	14	"	"	"	"	"
T11	3CS to 11B "	"	33,32	12	"	"	"	"	"
T12	3CS to 12B "	"	33,13	34	"	"	"	"	"
T13	3CS to 13B "	"	33,40	19	"	"	"	"	"
T14	3CS to 14B "	"	33,39	18	"	"	"	"	"
T15	3CS to 15B "	"	33,38	17	"	"	"	"	"
T16	3CS to 16B "	"	33,37	16	"	"	"	"	"
T17	1CS to 1A "	26	5	25	21	"	"	"	13,20,21
T18	1CS to 2A "	"	24	4	"	"	"	"	"
T19	1CS to 3A "	"	23	3	"	"	"	"	"
T20	1CS to 4A "	"	2	22	"	"	"	"	"
T21	2CS to 5A "	11	31	30	6	"	"	"	"

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TABLE XX. Output disable delay time (t_{HO}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T22	2CS to 6A Output disable delay time (t_{HO}), C_L = 50 pF	11	9	29	6	7	78	ns	13,20,21
T23	2CS to 7A "	"	8	28	"	"	"	"	"
T24	2CS to 8A "	"	7	27	"	"	"	"	"
T25	3CS to 9A "	20	15	36	33	"	"	"	"
T26	3CS to 10A "	"	14	35	"	"	"	"	"
T27	3CS to 11A "	"	12	32	"	"	"	"	"
T28	3CS to 12A "	"	34	13	"	"	"	"	"
T29	3CS to 13A "	"	19	40	"	"	"	"	"
T30	3CS to 14A "	"	18	39	"	"	"	"	"
T31	3CS to 15A "	"	17	38	"	"	"	"	"
T32	3CS to 16A "	"	16	37	"	"	"	"	"

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TABLE XXI. Output disable delay time (t_{1H}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1CS to 1B Output disable delay time (t_{1H}), C_L = 50 pF	26	21	5	25	4	45	ns	14,20,21
U2	1CS to 2B "	"	"	24	4	"	"	"	"
U3	1CS to 3B "	"	"	23	3	"	"	"	"
U4	1CS to 4B "	"	"	2	22	"	"	"	"
U5	2CS to 5B "	11	6	31	30	"	"	"	"
U6	2CS to 6B "	"	"	9	29	"	"	"	"
U7	2CS to 7B "	"	"	8	28	"	"	"	"
U8	2CS to 8B "	"	"	7	27	"	"	"	"
U9	3CS to 9B "	20	33	15	36	"	"	"	"
U10	3CS to 10B "	"	"	14	35	"	"	"	"
U11	3CS to 11B "	"	"	12	32	"	"	"	"
U12	3CS to 12B "	"	"	34	13	"	"	"	"
U13	3CS to 13B "	"	"	19	40	"	"	"	"
U14	3CS to 14B "	"	"	18	39	"	"	"	"
U15	3CS to 15B "	"	"	17	38	"	"	"	"
U16	3CS to 16B "	"	"	16	37	"	"	"	"
U17	1CS to 1A "	26	Open	25	21,5	10	220	"	13,20,21
U18	1CS to 2A "	"	"	4	21,24	"	"	"	"
U19	1CS to 3A "	"	"	3	21,23	"	"	"	"
U20	1CS to 4A "	"	"	22	21,2	"	"	"	"
U21	2CS to 5A "	11	"	30	6,31	"	"	"	"

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TABLE XXI. Output disable delay time (t_{1H}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U22	2CS to 6A Output disable delay time (t_{1H}), C_L = 50 pF	11	Open	29	6,9	10	220	ns	13,20,21
U23	2CS to 7A "	"	"	28	6,8	"	"	"	"
U24	2CS to 8A "	"	"	27	6,7	"	"	"	"
U25	3CS to 9A "	20	"	36	33,15	"	"	"	"
U26	3CS to 10A "	"	"	35	33,14	"	"	"	"
U27	3CS to 11A "	"	"	32	33,12	"	"	"	"
U28	3CS to 12A "	"	"	13	33,34	"	"	"	"
U29	3CS to 13A "	"	"	40	33,19	"	"	"	"
U30	3CS to 14A "	"	"	39	33,18	"	"	"	"
U31	3CS to 15A "	"	"	38	33,17	"	"	"	"
U32	3CS to 16A "	"	"	37	33,16	"	"	"	"

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TABLE XXII. Output disable delay time (t_{HI}) test.

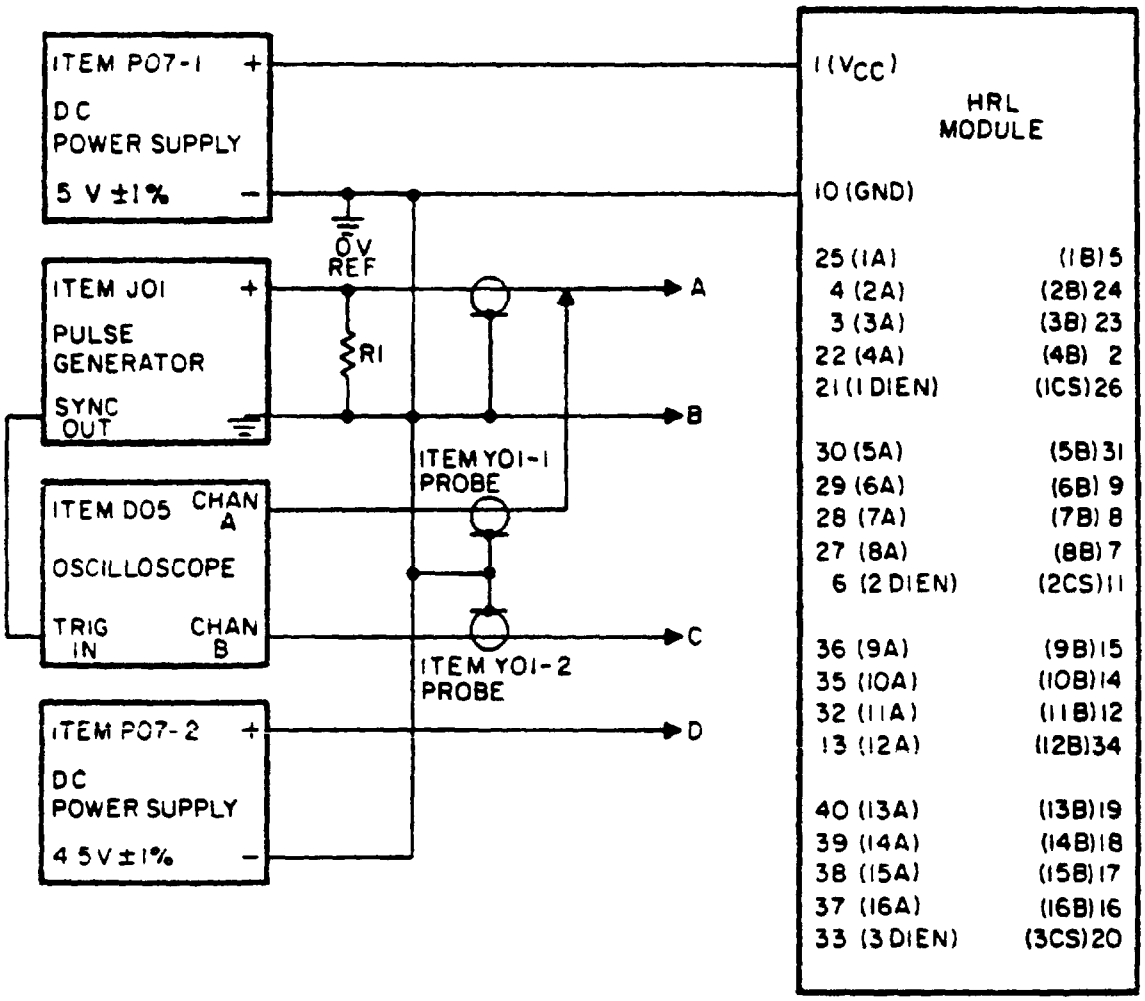
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1CS to 1B Output disable delay time (t_{HI}), C_L = 50 pF	26	21	5	25	7	78	ns	14,20,21
V2	1CS to 2B "	"	"	24	4	"	"	"	"
V3	1CS to 3B "	"	"	23	3	"	"	"	"
V4	1CS to 4B "	"	"	2	22	"	"	"	"
V5	2CS to 5B "	11	6	31	30	"	"	"	"
V6	2CS to 6B "	"	"	9	29	"	"	"	"
V7	2CS to 7B "	"	"	8	28	"	"	"	"
V8	2CS to 8B "	"	"	7	27	"	"	"	"
V9	3CS to 9B "	20	33	15	36	"	"	"	"
V10	3CS to 10B "	"	"	14	35	"	"	"	"
V11	3CS to 11B "	"	"	12	32	"	"	"	"
V12	3CS to 12B "	"	"	34	13	"	"	"	"
V13	3CS to 13B "	"	"	19	40	"	"	"	"
V14	3CS to 14B "	"	"	18	39	"	"	"	"
V15	3CS to 15B "	"	"	17	38	"	"	"	"
V16	3CS to 16B "	"	"	16	37	"	"	"	"
V17	1CS to 1A "	26	Open	25	21,5	"	"	"	13,20,21
V18	1CS to 2A "	"	"	4	21,24	"	"	"	"
V19	1CS to 3A "	"	"	3	21,23	"	"	"	"
V20	1CS to 4A "	"	"	22	21,2	"	"	"	"
V21	2CS to 5A "	11	"	30	6,31	"	"	"	"

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TABLE XXII. Output disable delay time (t_{HI}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V22	2CS to 6A Output disable delay time (t_{HI}), C_L = 50 pF	11	Open	29	6,9	7	78	ns	13,20,21
V23	2CS to 7A "	"	"	28	6,8	"	"	"	"
V24	2CS to 8A "	"	"	27	6,7	"	"	"	"
V25	3CS to 9A "	20	"	36	33,15	"	"	"	"
V26	3CS to 10A "	"	"	35	33,14	"	"	"	"
V27	3CS to 11A "	"	"	32	33,12	"	"	"	"
V28	3CS to 12A "	"	"	13	33,34	"	"	"	"
V29	3CS to 13A "	"	"	40	33,19	"	"	"	"
V30	3CS to 14A "	"	"	39	33,18	"	"	"	"
V31	3CS to 15A "	"	"	38	33,17	"	"	"	"
V32	3CS to 16A "	"	"	37	33,16	"	"	"	"

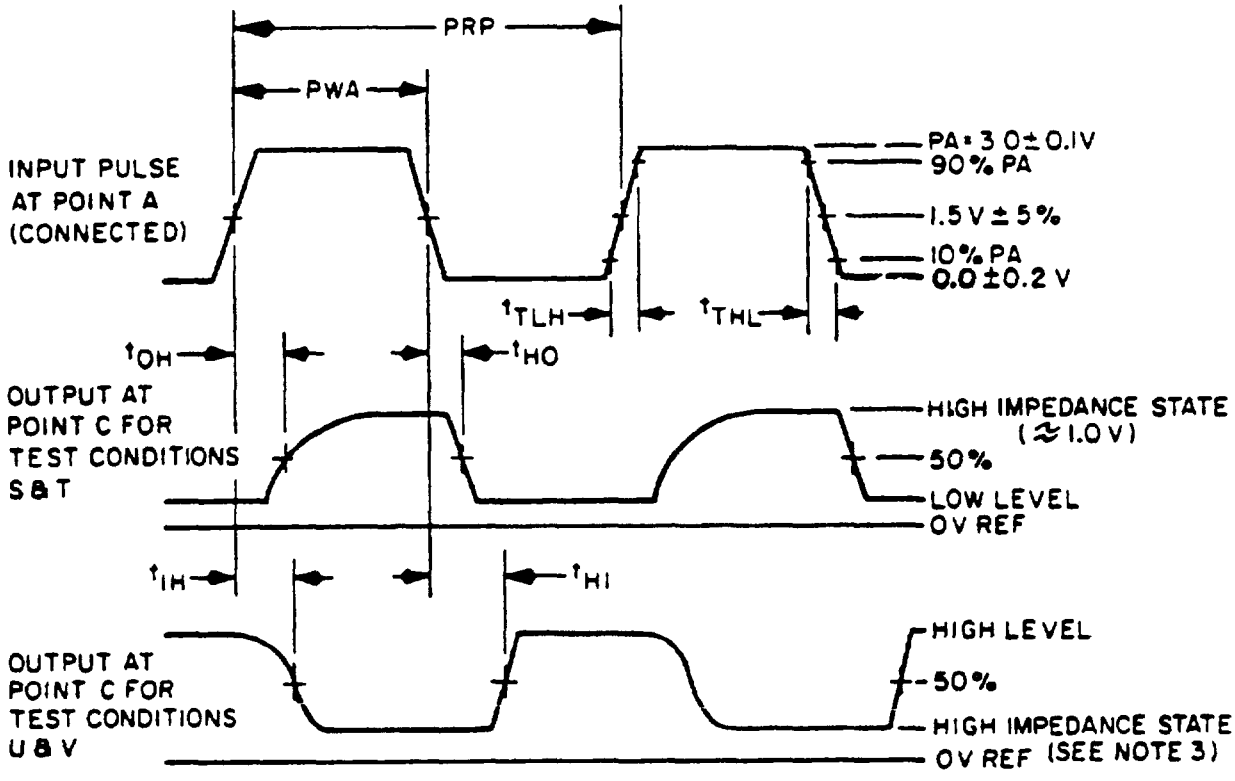
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Load output under test with load circuit as specified in test figure column of tables XIX, XX, XXI, and XXII.

FIGURE 20 Output disable delay time (t_{OH}, t_{HO}, t_{1H}, and t_{H1}) test circuit for tables XIX, XX, XXI, and XXII.

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NOTES
1. Input pulse setup
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
2. Before making measurements, verify that the output high and low levels are ≥2.4 V and ≤0.4 V for test conditions S1-S16, T1-T16, U1-U16, and V1-V16 and the high and low levels are ≥3.4 V and ≤ 0.45 V respectively for test conditions S17-S32, T17-T32, U17-U32, and V17-V32
3. For test conditions U1-U16 and V1-V16 the high impedance state voltage is ≈2 V, while for test conditions U17-U32 and V17-V32 the high impedance state voltage is ≈3 V.
4. All voltage levels are referenced to ground

FIGURE 21. Output disable delay time (t_{OH}, t_{HO}, t_{IH}, and t_{HI}) test circuit waveforms for tables XIX, XX, XXI, and XXII

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TABLE XXIII. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	1B Output crosstalk, low level	Open	25,21,26, 6,11,33,20	4,3,22,30,29,28,27,36,35,32, 13,40,39,38,37	5	0	1,200	mV	14,22,23
W2	2B "	"	4,21,26,6, 11,33,20	25,3,22,30,29,28,27,36,35,32, 13,40,39,38,37	24	"	"	"	"
W3	3B "	"	3,21,26,6, 11,33,20	25,4,22,30,29,28,27,36,35,32, 13,40,39,38,37	23	"	"	"	"
W4	4B "	"	22,21,26, 6,11,33,20	25,4,3,30,29,28,27,36,35,32, 13,40,39,38,37	2	"	"	"	"
W5	5B "	"	30,21,26, 6,11,33,20	25,4,3,22,29,28,27,36,35,32, 13,40,39,38,37	31	"	"	"	"
W6	6B "	"	29,21,26, 6,11,33,20	25,4,3,22,30,28,27,36,35,32, 13,40,39,38,37	9	"	"	"	"
W7	7B "	"	28,21,26, 6,11,33,20	25,4,3,22,30,29,27,36,35,32, 13,40,39,38,37	8	"	"	"	"
W8	8B "	"	27,21,26, 6,11,33,20	25,4,3,22,30,29,28,36,35,32, 13,40,39,38,37	7	"	"	"	"
W9	9B "	"	36,21,26, 6,11,33,20	25,4,3,22,30,29,28,27,35,32, 13,40,39,38,37	15	"	"	"	"
W10	10B "	"	35,21,26, 6,11,33,20	25,4,3,22,30,29,28,27,36,32, 13,40,39,38,37	14	"	"	"	"
W11	11B "	"	32,21,26, 6,11,33,20	25,4,3,22,30,29,28,27,36,35, 13,40,39,38,37	12	"	"	"	"
W12	12B "	"	13,21,26, 6,11,33,20	25,4,3,22,30,29,28,27,36,35, 32,40,39,38,37	34	"	"	"	"
W13	13B "	"	40,21,26, 6,11,33,20	25,4,3,22,30,29,28,27,36,35, 32,13,39,38,37	19	"	"	"	"
W14	14B "	"	39,21,26, 6,11,33,20	25,4,3,22,30,29,28,27,36,35, 32,13,40,38,37	18	"	"	"	"

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TABLE XXIII. Output crosstalk (low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
W15	15B	Output crosstalk, low level	Open	38,21,26, 6,11,33,20	25,4,3,22,30,29,28,27,36,35, 32,13,40,39,37	17	0	1,200	mV	14,22,23
W16	16B	"	Open	37,21,26, 6,11,33,20	25,4,3,22,30,29,28,27,36,35, 32,13,40,39,38	16	"	"	"	14,22,23
W17	1A	"	21,6,33	5,26, 11,20	24,23,2,31,9,8,7,15,14,12,34, 19,18,17,16	25	"	"	"	13,22,23
W18	2A	"	"	24,26, 11,20	5,23,2,31,9,8,7,15,14,12,34, 19,18,17,16	4	"	"	"	"
W19	3A	"	"	23,26, 11,20	5,24,2,31,9,8,7,15,14,12,34, 19,18,17,16	3	"	"	"	"
W20	4A	"	"	2,26, 11,20	5,24,23,31,9,8,7,15,14,12,34, 19,18,17,16	22	"	"	"	"
W21	5A	"	"	31,26, 11,20	5,24,23,2,9,8,7,15,14,12,34, 19,18,17,16	30	"	"	"	"
W22	6A	"	"	9,26, 11,20	5,24,23,2,31,8,7,15,14,12,34, 19,18,17,16	29	"	"	"	"
W23	7A	"	"	8,26, 11,20	5,24,23,2,31,9,7,15,14,12,34, 19,18,17,16	28	"	"	"	"
W24	8A	"	"	7,26, 11,20	5,24,23,2,31,9,8,15,14,12,34, 19,18,17,16	27	"	"	"	"
W25	9A	"	"	15,26, 11,20	5,24,23,2,31,9,8,7,14,12,34, 19,18,17,16	36	"	"	"	"
W26	10A	"	"	14,26, 11,20	5,24,23,2,31,9,8,7,15,12,34, 19,18,17,16	35	"	"	"	"
W27	11A	"	"	12,26, 11,20	5,24,23,2,31,9,8,7,15,14,34, 19,18,17,16	32	"	"	"	"

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TABLE XXIII Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W28	12A Output crosstalk, low level	21,6,33	34,26, 11,20	5,24,23,2,31,9,8,7,15,14,12, 19,18,17,16	13	0	1,200	mV	13,22,23
W29	13A "	"	19,26, 11,20	5,24,23,2,31,9,8,7,15,14,12, 34,18,17,16	40	"	"	"	"
W30	14A "	"	18,26, 11,20	5,24,23,2,31,9,8,7,15,14,12, 34,19,17,16	39	"	"	"	"
W31	15A "	"	17,26, 11,20	5,24,23,2,31,9,8,7,15,14,12, 34,19,18,16	38	"	"	"	"
W32	16A "	"	16,26, 11,20	5,24,23,2,31,9,8,7,15,14,12, 34,19,18,17	37	"	"	"	"

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TABLE XXIV. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	1B Output crosstalk, high level	25	21,26,6, 11,33,20	4,3,22,30,29,28,27,36,35,32, 13,40,39,38,37	5	0	800	mV	14,22,23
X2	2B "	4	"	25,3,22,30,29,28,27,36,35,32, 13,40,39,38,37	24	"	"	"	"
X3	3B "	3	"	25,4,22,30,29,28,27,36,35,32, 13,40,39,38,37	23	"	"	"	"
X4	4B "	22	"	25,4,3,30,29,28,27,36,35,32, 13,40,39,38,37	2	"	"	"	"
X5	5B "	30	"	25,4,3,22,29,28,27,36,35,32, 13,40,39,38,37	31	"	"	"	"
X6	6B "	29	"	25,4,3,22,30,28,27,36,35,32, 13,40,39,38,37	9	"	"	"	"
X7	7B "	28	"	25,4,3,22,30,29,27,36,35,32, 13,40,39,38,37	8	"	"	"	"
X8	8B "	27	"	25,4,3,22,30,29,28,36,35,32, 13,40,39,38,37	7	"	"	"	"
X9	9B "	36	"	25,4,3,22,30,29,28,27,35,32, 13,40,39,38,37	15	"	"	"	"
X10	10B "	35	"	25,4,3,22,30,29,28,27,36,32, 13,40,39,38,37	14	"	"	"	"
X11	11B "	32	"	25,4,3,22,30,29,28,27,36,35, 13,40,39,38,37	12	"	"	"	"
X12	12B "	13	"	25,4,3,22,30,29,28,27,36,35, 32,40,39,38,37	34	"	"	"	"
X13	13B "	40	"	25,4,3,22,30,29,28,27,36,35, 32,13,39,38,37	19	"	"	"	"
X14	14B "	39	"	25,4,3,22,30,29,28,27,36,35, 32,13,40,38,37	18	"	"	"	"

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TABLE XXIV. Output crosstalk (high-level) test - Continued.

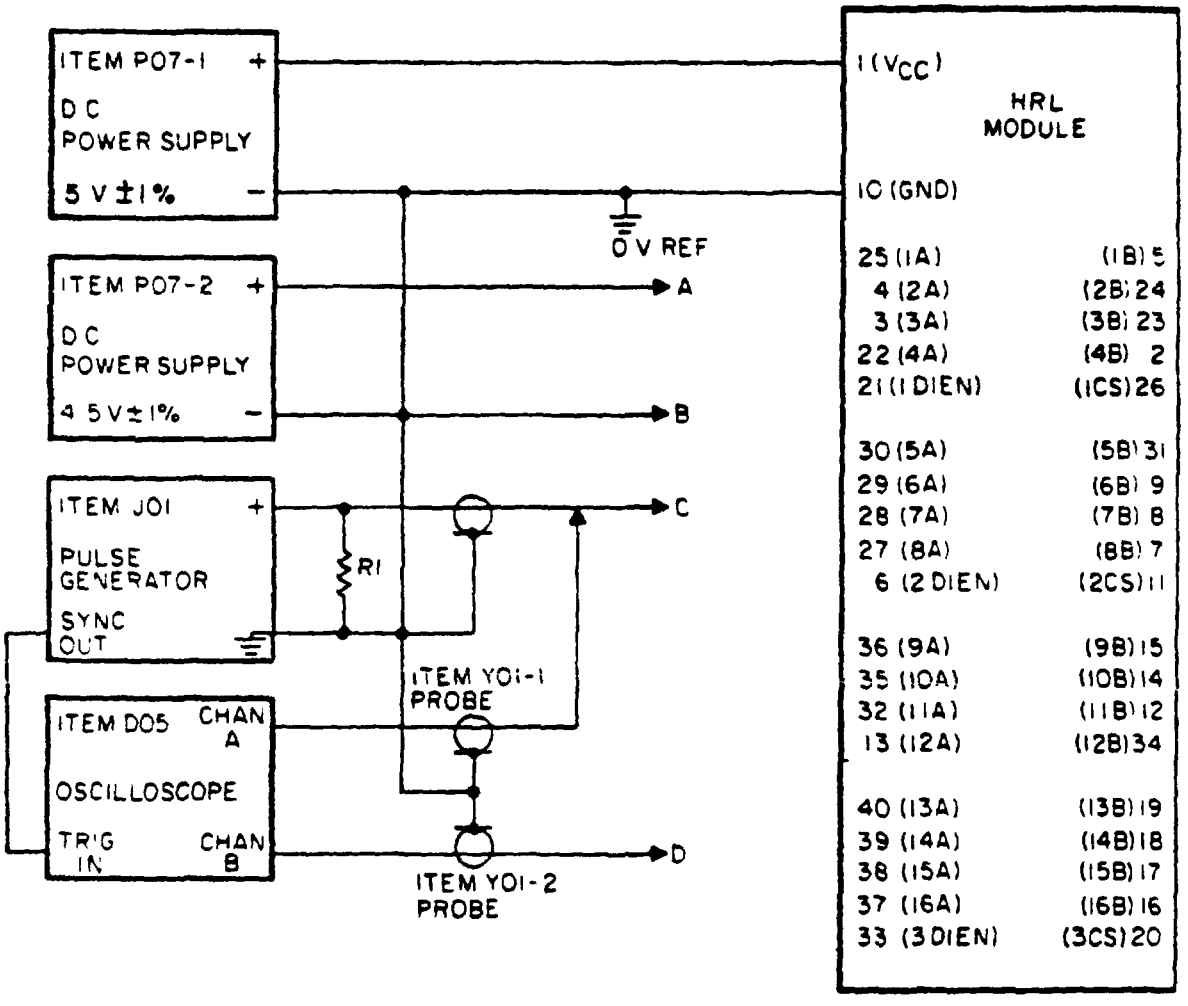
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X15	15B Output crosstalk, high level	38	21,26,6, 11,33,20	25,4,1,22,30,29,28,27,36,35, 32,13,40,39,37	17	0	800	mV	14,22,23
X16	16B "	37	21,26,6, 11,33,20	25,4,1,22,30,29,28,27,36,35, 32,13,40,39,38	16	"	"	"	14,22,23
X17	1A "	5,21,6,33	26,11,20	24,23,2,31,9,8,7,15,14,12, 34,19,18,17,16	25	"	"	"	13,22,23
X18	2A "	24,21,6,33	"	5,23,2,31,9,8,7,15,14,12, 34,19,18,17,16	4	"	"	"	"
X19	3A "	23,21,6,33	"	5,24,2,31,9,8,7,15,14,12, 34,19,18,17,16	3	"	"	"	"
X20	4A "	2,21,6,33	"	5,24,23,31,9,8,7,15,14,12, 34,19,18,17,16	22	"	"	"	"
X21	5A "	31,21,6,33	"	5,24,23,2,9,8,7,15,14,12, 34,19,18,17,16	30	"	"	"	"
X22	6A "	9,21,6,33	"	5,24,23,2,31,8,7,15,14,12, 34,19,18,17,16	29	"	"	"	"
X23	7A "	8,21,6,33	"	5,24,23,2,31,9,7,15,14,12, 34,19,18,17,16	28	"	"	"	"
X24	8A "	7,21,6,33	"	5,24,23,2,31,9,8,15,14,12, 34,19,18,17,16	27	"	"	"	"
X25	9A "	15,21,6,33	"	5,24,23,2,31,9,8,7,14,12, 34,19,18,17,16	36	"	"	"	"
X26	10A "	14,21,6,33	"	5,24,23,2,31,9,8,7,15,12, 34,19,18,17,16	35	"	"	"	"
X27	11A "	12,21,6,33	"	5,24,23,2,31,9,8,7,15,14, 34,19,18,17,16	32	"	"	"	"
X28	12A "	34,21,6,33	"	5,24,23,2,31,9,8,7,15,14, 12,19,18,17,16	13	"	"	"	"

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TABLE XXIV Output crosstalk (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X29	13A Output crosstalk, high level	19,21,6,33	26,11,20	5,24,23,2,31,9,8,7,15,14, 12,34,18,17,16	40	0	800	mV	13,22,23
X30	14A "	18,21,6,33	"	5,24,23,2,31,9,8,7,15,14, 12,34,19,17,16	39	"	"	"	"
X31	15A "	17,21,6,33	"	5,24,23,2,31,9,8,7,15,14, 12,34,19,18,16	38	"	"	"	"
X32	16A "	16,21,6,33	"	5,24,23,2,31,9,8,7,15,14, 12,34,19,18,17	37	"	"	"	"

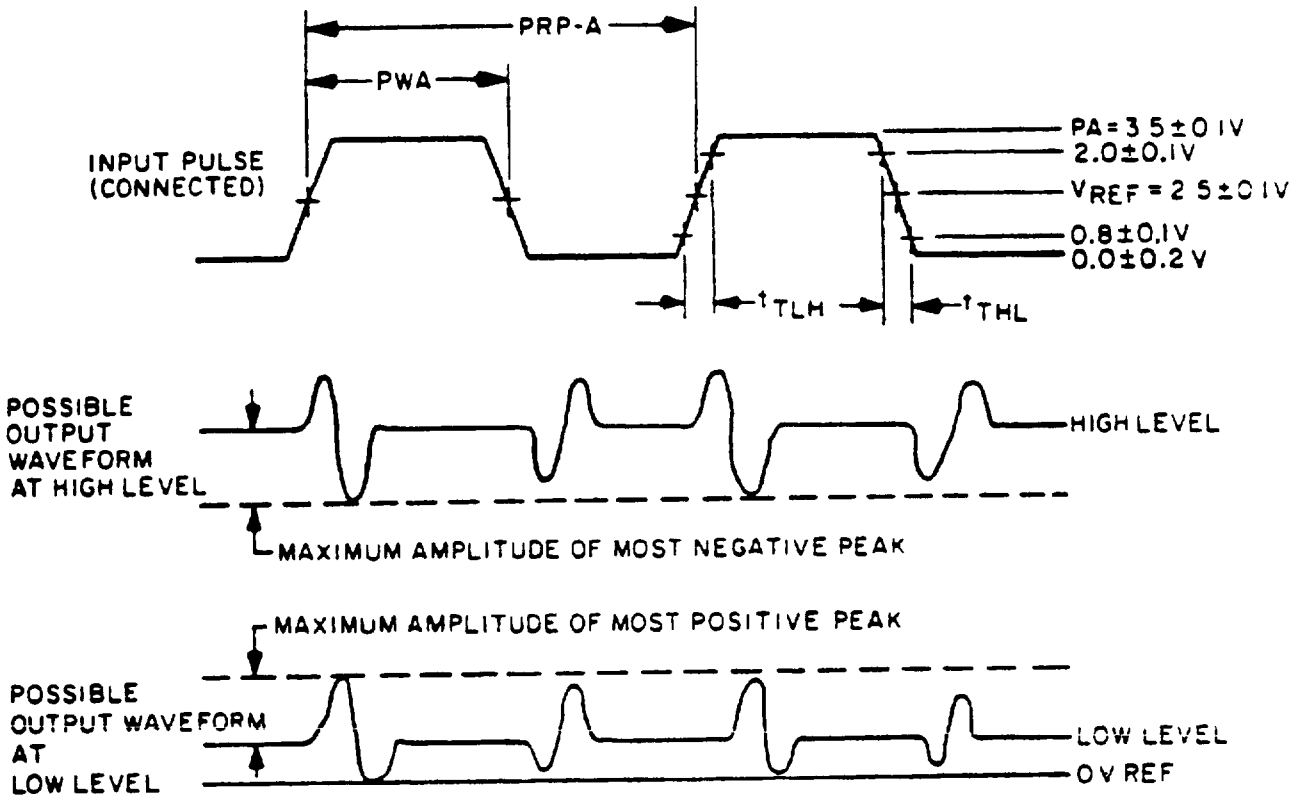
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NOTES.
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Load all outputs with load as specified in test figure column of tables XXIII and XXIV.

FIGURE 22. Output crosstalk (low-level and high-level) test circuit for tables XXIII and XXIV.

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NOTES.

1. Input pulse parameters. For each test condition with the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude, pulse low level voltage, and transition times. Verify the parameters at each input pin before making each measurement.

   t_{TLH} = 7.5 ±1.5 ns (verified at each input pin).
   PWA = 500 ±50 ns
   PRP = 1.0 ±0.1 us.
   t_{THL} = 13.0 ±2.0 ns (verified at each input pin).

2. Before making the measurements specified in tables XXIII and XXIV, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤0.45 V for test conditions W1-W16 and X1-X16 and that the output pulse high level is ≥ 3.4 V and the low level is ≤ 0.45 V for test conditions W17-W32 and X17-X32.

FIGURE 23. Output crosstalk (low-level and high-level) test circuit waveforms for tables XXIII and XXIV.

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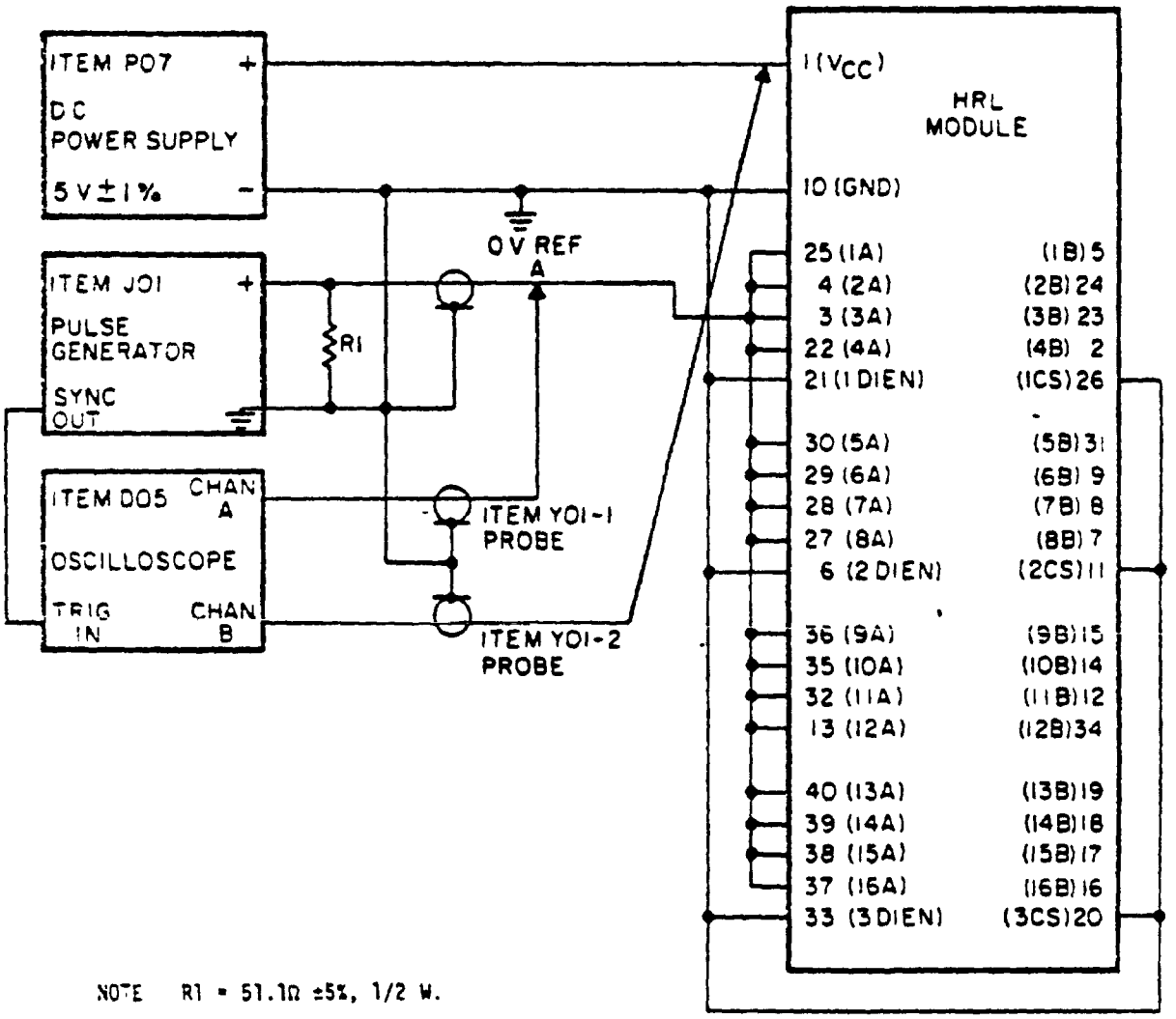
TABLE XXV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient amplitude	0	800	mV	24,25

TABLE XXVI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z	Power supply filter capacitance	1	Open	0.71	1.31	µF	26

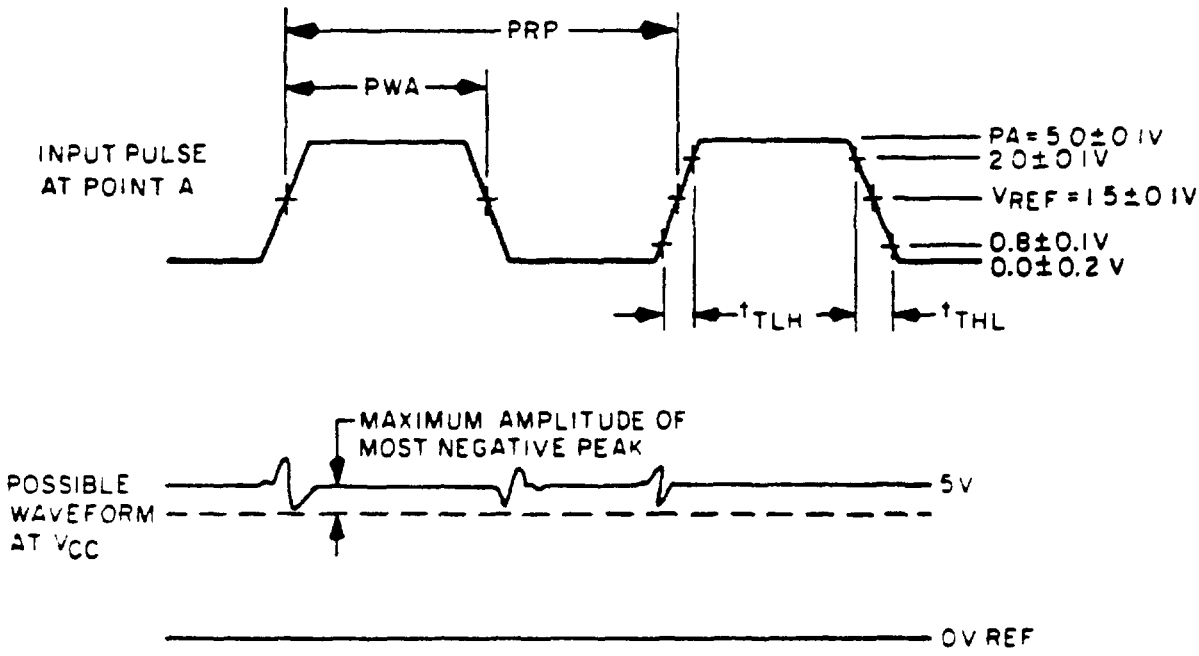
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NOTE R1 = 51.1Ω ±5%, 1/2 W.

FIGURE 24 Power supply transient amplitude test circuit for table XXV

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NOTE Input pulse setup:
PWA = 500 ±50 ns.
PRP = 1.0 ±0.1 µs.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

FIGURE 25. Power supply transient amplitude test circuit waveforms for table XXV

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TABLE XXVII. Input capacitance test.

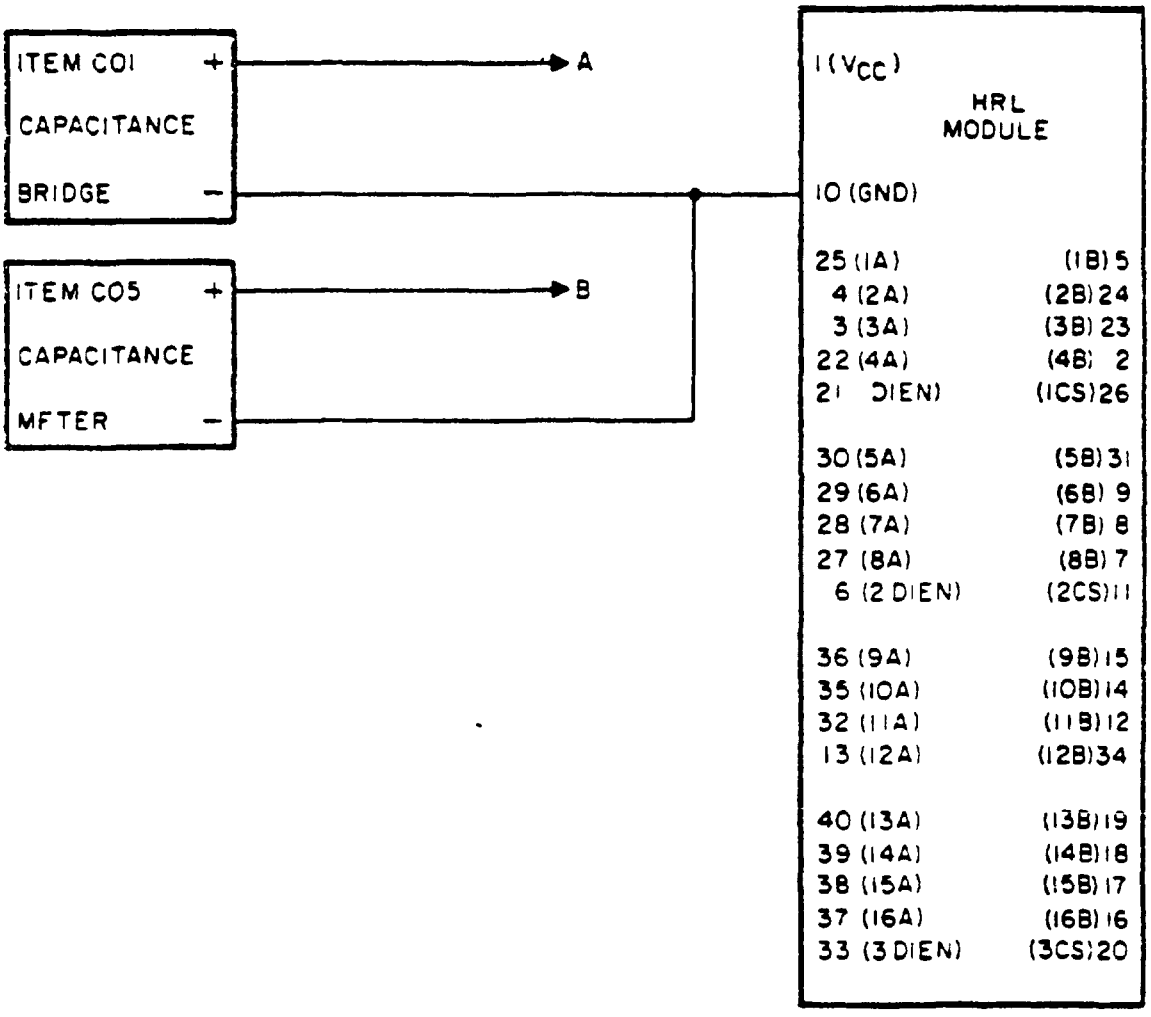
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA1	1A Input capacitance	Open	25	2	25	pF	26
AA2	2A "	"	4	"	"	"	"
AA3	3A "	"	3	"	"	"	"
AA4	4A "	"	22	"	"	"	"
AA5	5A "	"	30	"	"	"	"
AA6	6A "	"	29	"	"	"	"
AA7	7A "	"	28	"	"	"	"
AA8	8A "	"	27	"	"	"	"
AA9	9A "	"	36	"	"	"	"
AA10	10A "	"	35	"	"	"	"
AA11	11A "	"	32	"	"	"	"
AA12	12A "	"	13	"	"	"	"
AA13	13A "	"	40	"	"	"	"
AA14	14A "	"	39	"	"	"	"
AA15	15A "	"	38	"	"	"	"
AA16	16A "	"	37	"	"	"	"
AA17	1B "	"	5	"	"	"	"
AA18	2B "	"	24	"	"	"	"
AA19	3B "	"	23	"	"	"	"
AA20	4B "	"	2	"	"	"	"
AA21	5B "	"	31	"	"	"	"

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TABLE XXVII Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA22	6B Input capacitance	Open	9	2	25	pF	26
AA23	7B "	"	8	"	"	"	"
AA24	8B "	"	7	"	"	"	"
AA25	9B "	"	15	"	"	"	"
AA26	10B "	"	14	"	"	"	"
AA27	11B "	"	12	"	"	"	"
AA28	12B "	"	34	"	"	"	"
AA29	13B "	"	19	"	"	"	"
AA30	14B "	"	18	"	"	"	"
AA31	15B "	"	17	"	"	"	"
AA32	16B "	"	16	"	"	"	"
AA33	1DIEN "	"	21	"	15	"	"
AA34	1CS "	"	26	"	"	"	"
AA35	2DIEN "	"	6	"	"	"	"
AA36	2CS "	"	11	"	"	"	"
AA37	3DIEN "	"	33	4	30	"	"
AA38	3CS "	"	20	4	30	"	"

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NOTES
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins under test. Stray capacitance added to other pins may affect the readings
3. Capacitance bridge (C01) setup
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 26 Power supply filter and input capacitance test circuit for tables XXVI and XXVII

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